



Rabbits and Chaos

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AY 2025-2026

Informatica 2.2 / Computational Intelligence

Outline

- Rabbits
- Fibonacci and the Golden Ratio
- Population models
 - Malthus
 - Verhulst
 - Logistic map
- Chaos
- Analysis of chaotic systems
- Chaos and fractals
- Learning Chaos: Echo-State Networks

Fibonacci sequence

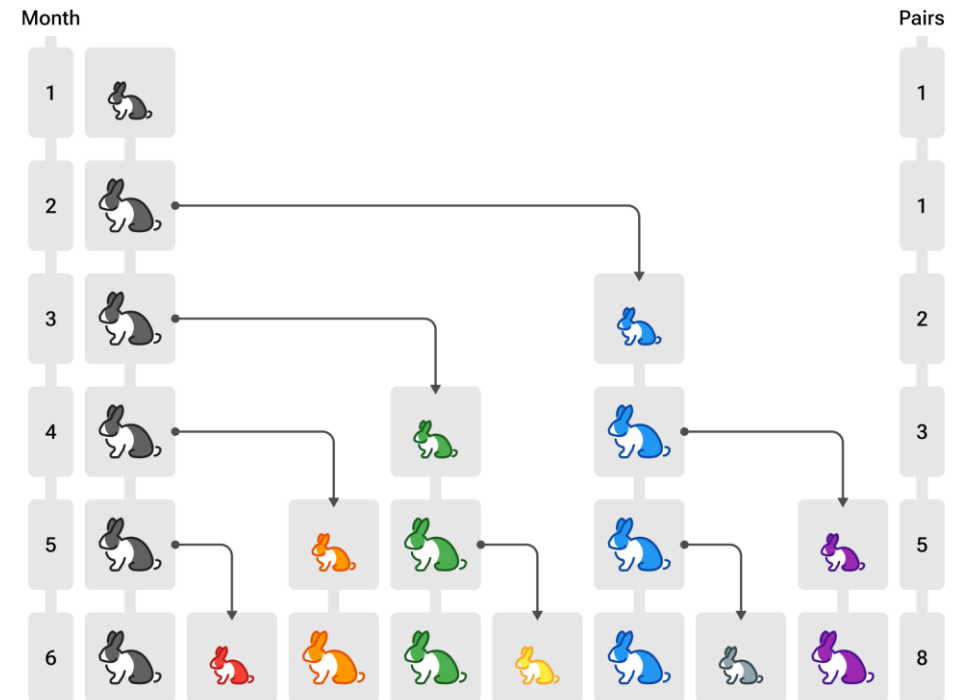
- **In 1202**, mathematician Leonardo «dei Bonacci» (aka **Fibonacci**) published the book *Liber Abbaci*
 - It was a textbook about arithmetic and algebra
 - That introduced the **indo-arabic decimal system** in Europe
 - He also introduced new concepts, methods, and **problems**
- One of the problems proposed by Fibonacci was related to the **growth of a rabbits population**



Rabbits..?

Fibonacci considers the idealized (hence, not so realistic) growth based on the following assumptions:

1. A rabbit **becomes mature** after one month
 2. A rabbit **gives birth after the second month**
 3. Rabbits **never die**
- At month 1 there is only **1** rabbit
 - At month 2 there is only **1** (mature) rabbit
 - At month 3 there are **2** rabbits: the first one and the **child** (not mature)
 - At month 4 there are **3** rabbits: the first one, the first **child** (now mature), and a new **child** (not yet mature)
 - If we **count the rabbits** during each month, we see the following **sequence**: 1, 1, 2, 3, 5, 8, 13, ...



Fibonacci's sequence

- 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...
- If you inspect the sequence closely, **each number in the sequence is the sum of the previous two** (except the first two...)
- Thus, we define the Fibonacci sequence as a **recursive formula**:

$$F(n) = F(n - 1) + F(n - 2)$$

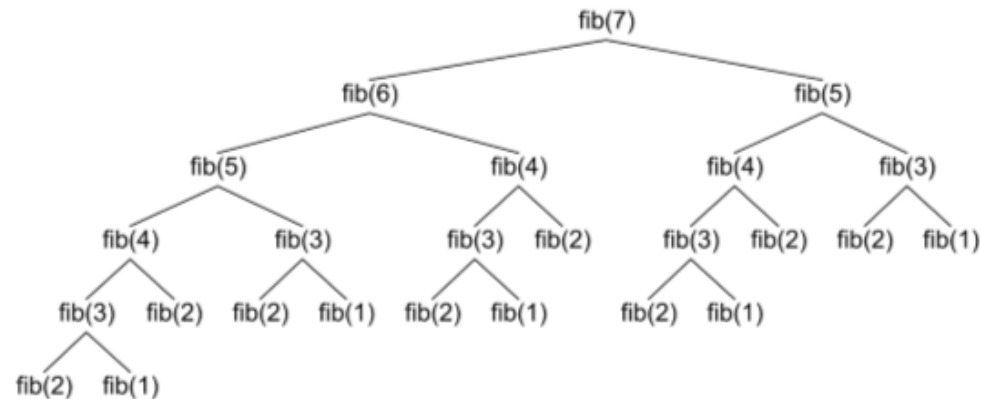
$$F(0) = 0$$

$$F(1) = 1$$

Fibonacci's sequence is not trivial

$$F(n) = F(n - 1) + F(n - 2)$$

- Can you tell me the value for $F(20)$? That's hard, right?
- Especially because it is a recursive formula that involves **two Fibonacci sub-sequences**, the complexity is exponential!



Closed formula and golden ratio

- Believe it or not, in **1843 Binet published a closed-form formula** to calculate any Fibonacci number

$$F(n) = \frac{\varphi^n - \psi^n}{\sqrt{5}}$$

where $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$

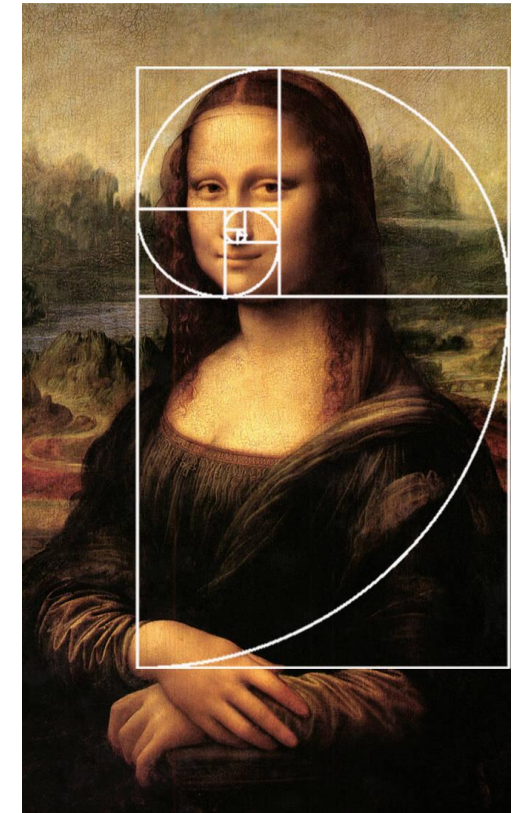
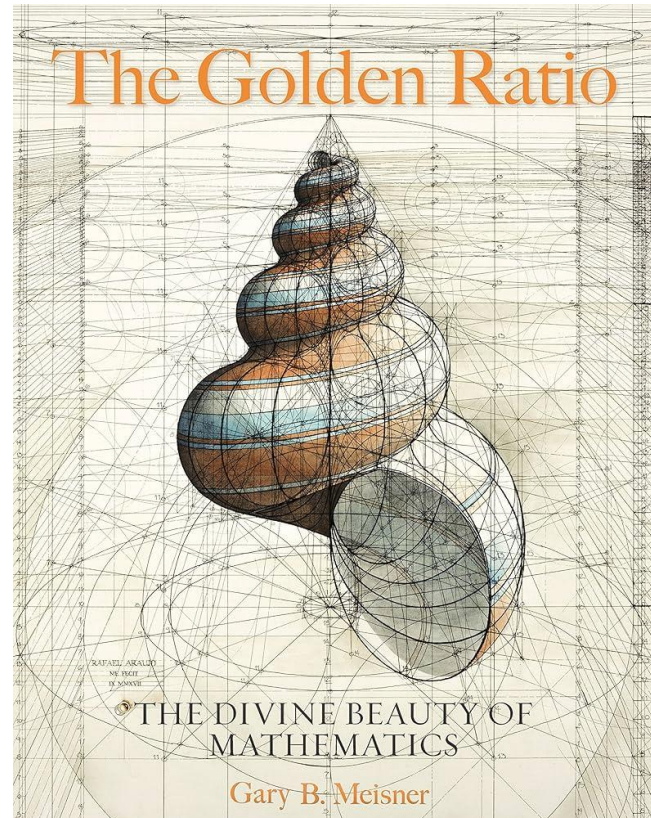
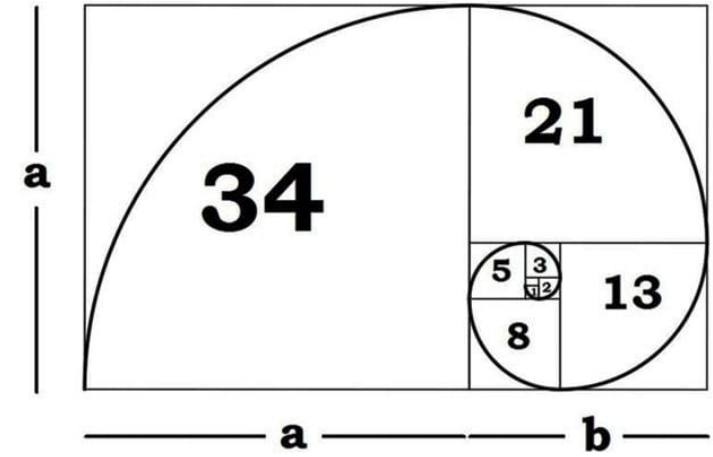
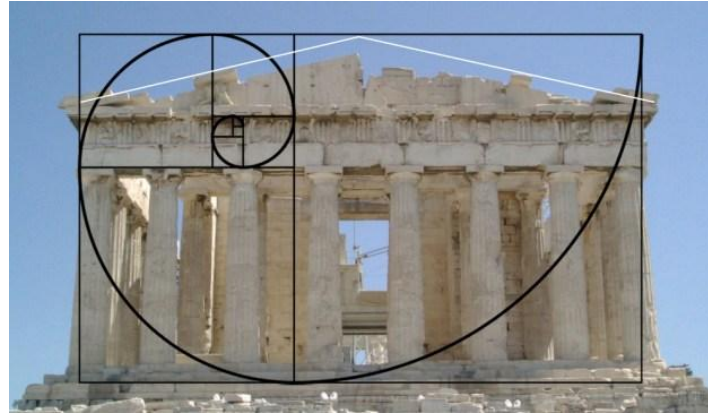
Applying Binet's formula

$$F(20) = \frac{\varphi^{20} - \psi^{20}}{\sqrt{5}} = \frac{\left(\frac{1 + \sqrt{5}}{2}\right)^{20} - \left(\frac{1 - \sqrt{5}}{2}\right)^{20}}{\sqrt{5}} = \mathbf{6575}$$

φ is known as the **Golden Ratio**

1
1 (0 + 1)
2 (1 + 1)
3 (1 + 2)
5 (2 + 3)
8 (3 + 5)
13 (5 + 8)
21 (8 + 13)
34 (13 + 21)
55 (21 + 34)
89 (34 + 55)
144 (55 + 89)
233 (89 + 144)
377 (144 + 233)
610 (233 + 377)
987 (377 + 610)
1597 (610 + 987)
2584 (987 + 1597)
4181 (1597 + 2584)
6765 (2584 + 4181) 

Golden ratio?





Population growth models

- The Fibonacci sequence is one “toy” example of population growth model
- Late 18th-century biologists began to **develop real models and techniques** to understand the **dynamics** of growing and shrinking populations of living organisms
- One important milestone: the work of **Thomas Malthus**

Malthusian growth model

- Malthus wrote the book «An Essay on the Principle of Population» (1798)
 - One of the earliest and most influential books on that topic
 - **Simple exponential law** to calculate the population at time t

$$P(t) = P_0 e^{rt}$$

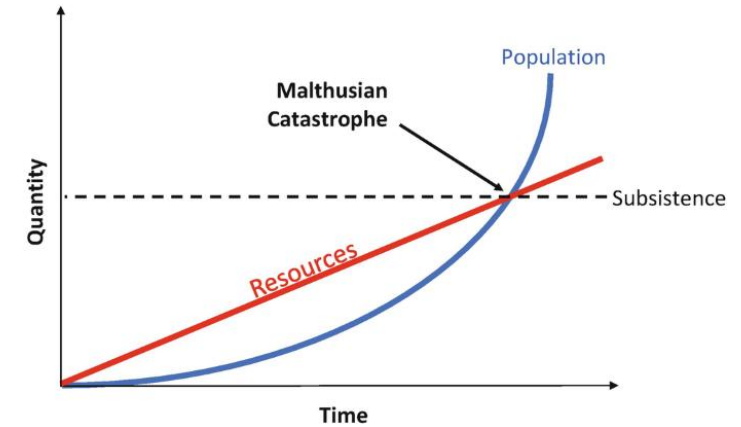
where P_0 is the initial population and r is the growth rate



- In population ecology, this is called «the first principle of population dynamics». However, it has a *tiny* problem... do you see it?

Infinite growth?

- In Malthus' model **human population** grows **exponentially**, but **resources** do not!
- Food production follows a more **arithmetical** growth, so that eventually the **population oustrips resources** leading to collapse



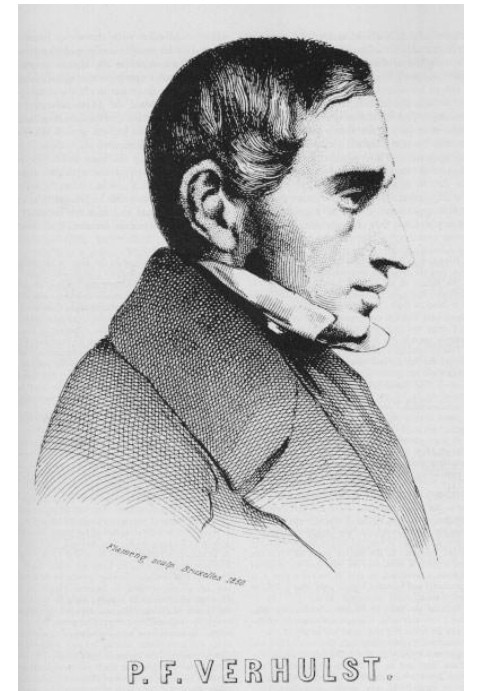
Check out [\[Burger, Encyclopedia of Evolutionary Psychological Science \(4751:4760\), 2021\]](#)

Verhulst

- **Pierre Franois Verhulst** proposed in 1838 a **new version** of the model, this time **bounded by resource limitations**
- It is called the «**logistic equation**»:

$$\frac{dN}{dt} = rN - \alpha N^2$$

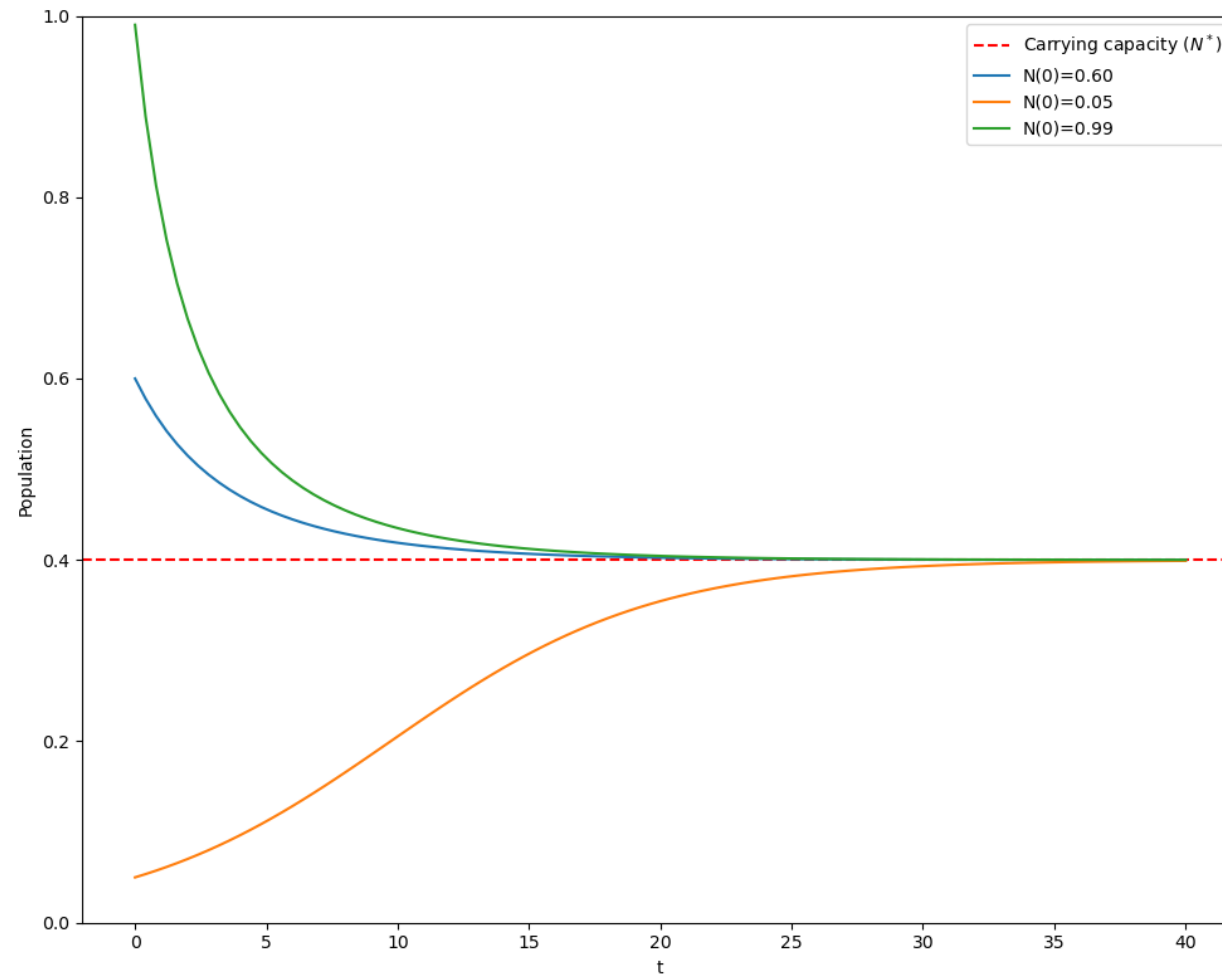
where the variable N is the number of individuals, r is the usual **growth rate**, and α is a new parameter representing **crowding/competition** for resources



The logistic equation

- One important concept of the logistic equation is the «**carrying capacity**», which corresponds to the population **equilibrium**
- That value corresponds to $N^* = \frac{r}{\alpha}$
- The curve **always converges to that value** (that is, it is a «global attractor» of the dynamical system) – let's see that in action

Attraction to the carrying capacity



Maximum number of individuals

- Pearl and Reed proposed an alternative logistic equation, in which $\alpha = \frac{r}{K}$ where K is the **maximum number of individuals** an environment can support
- By substituting and rearranging the terms, we get

$$\frac{dN}{dt} = rN - \alpha N^2 = rN - \frac{r}{K} N^2 = rN \left(1 - \frac{N}{K} \right)$$

Logistic map

- Let us now consider a **discrete-time version** of the logistic equation ranging in $[0,1]$. So instead of

$$rN \left(1 - \frac{N}{K} \right)$$

let us write

$$N_{t+1} = rN_t(1 - N_t)$$

- This **nonlinear difference equation** is called the **logistic map**

What is a map anyway?

- An **iterative map** is a mathematical function that is repeatedly applied to its own output

$$x_{t+1} = f(x_t)$$

- Iterative maps are used in many fields and disciplines to study the **dynamical properties** of the system over time, and can have a very **rich and unpredictable behavior**
- If you have a *deja-vu* (e.g., cellular automata), that's a good signal

The meaning of the Logistic map

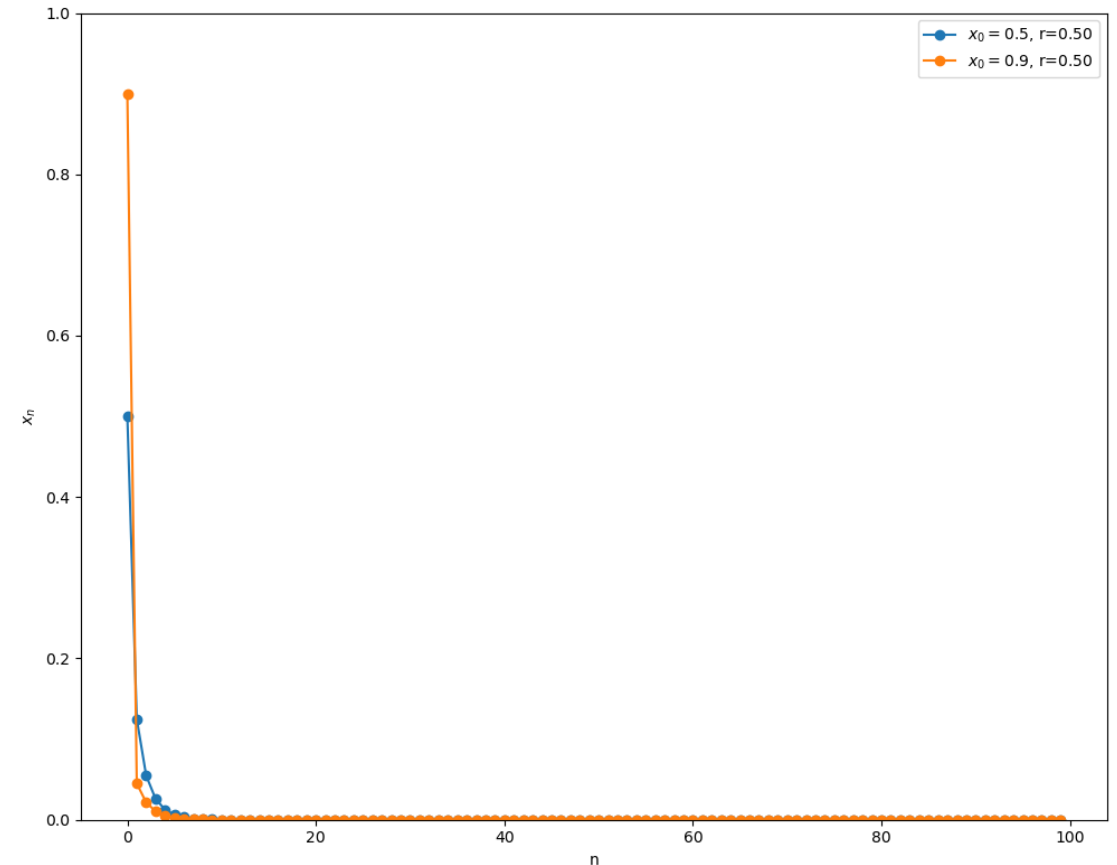
- In the literature, a logistic map is usually written as $x_{n+1} = rx_n(1 - x_n)$
- $x_n \in (0,1)$ represents the ratio of existing population to the maximum possible population
- $r \in [0,4]$ because smaller values do not make sense while greater values lead to negative populations (which also make no sense)
- The equation (once again) models two phenomena:
 - **Reproduction** (the population grows proportionally to the current one)
 - **Starvation** (the population shrinks according to the carrying capacity)

Behavior of the logistic map: $r=0.5$

- Let's play with the recurrence equation

$$x_{n+1} = rx_n(1 - x_n)$$

- Let's assume $r=0.5$
- Regardless of the initial value,
the population always dies
- This is true for any value of r in the interval **[0,1]**

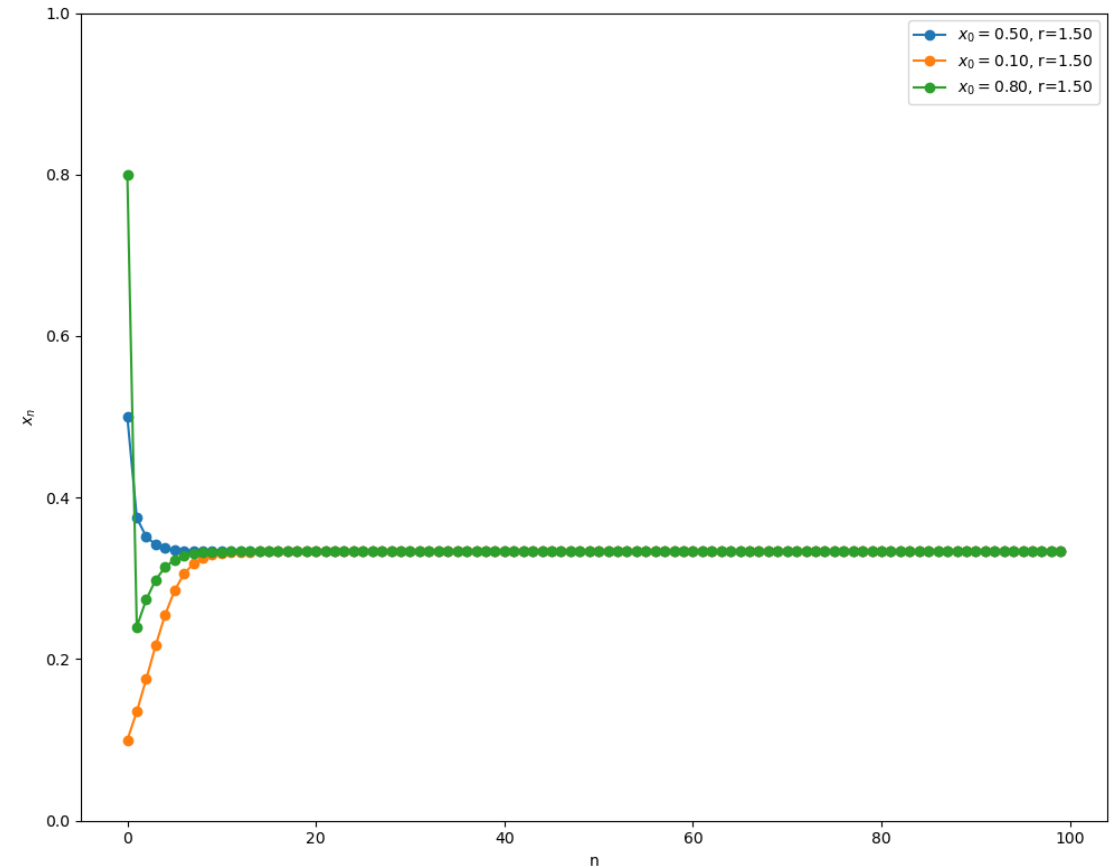


Behavior of the logistic map: $r=1.5$

- $x_{n+1} = rx_n(1 - x_n)$
- Let's consider $r=1.5$ this time
- Regardless of the initial value, **the population is always attracted to**

$$x = \frac{r - 1}{r} = 0.33 \dots$$

- This is true for any value of r in the interval **[1,2]**

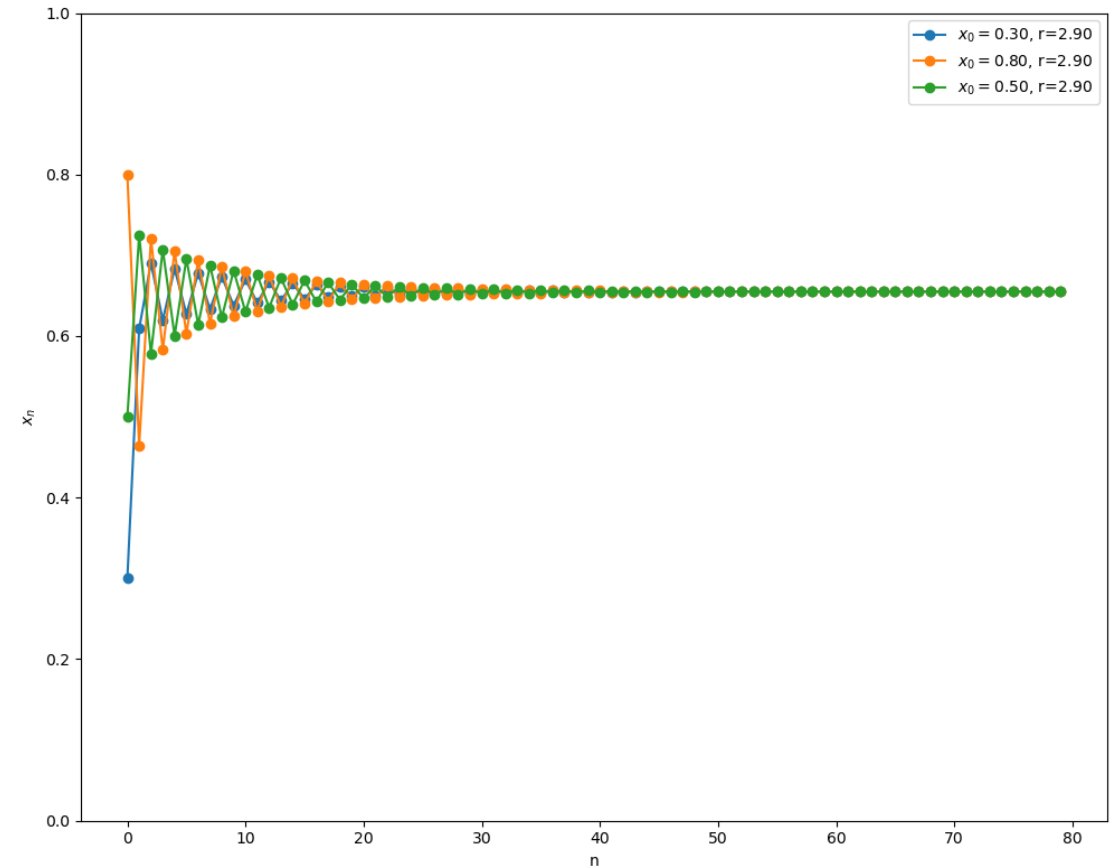


Behavior of the logistic map: $r=2.9$

- $x_{n+1} = rx_n(1 - x_n)$
- Let's consider $r=2.9$ this time
- Regardless of the initial value, **the population is again attracted to**

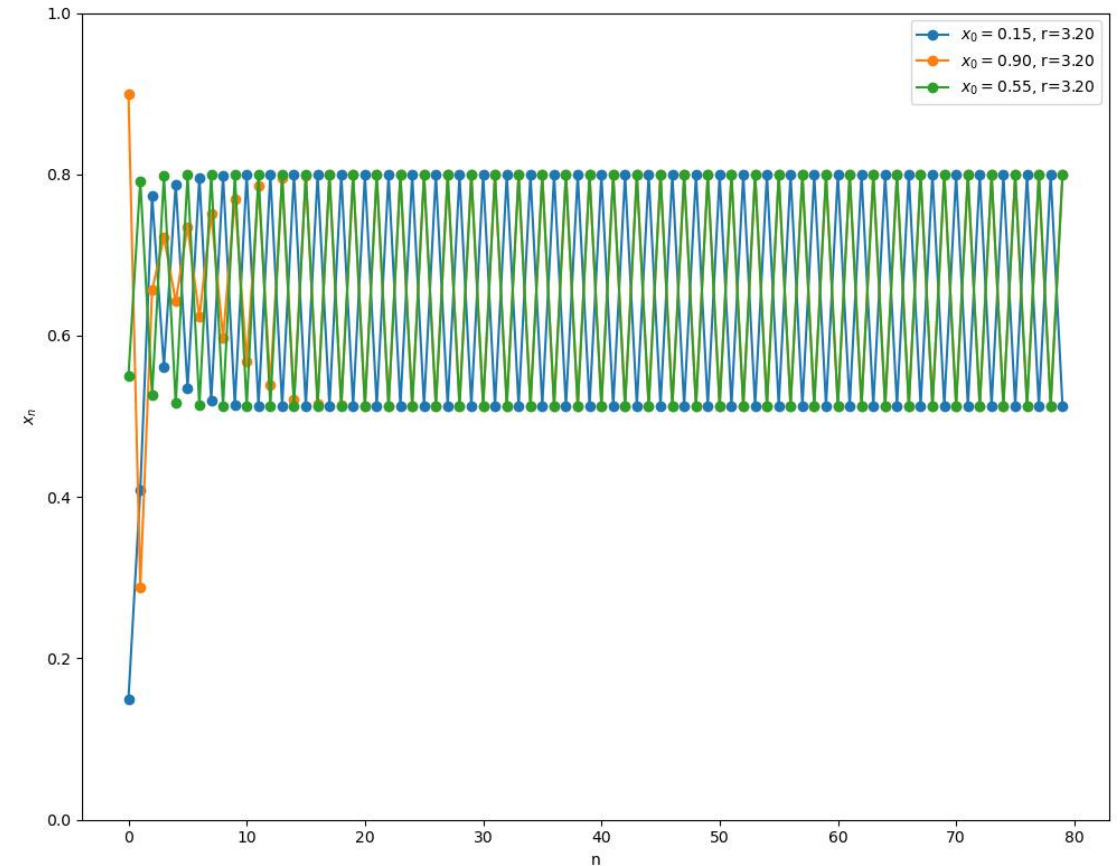
$$x = \frac{r-1}{r} = 0,65 \dots$$

- ..but it shows damped **oscillations around the attractor**
- True for all values of r in the interval **[2,3]**



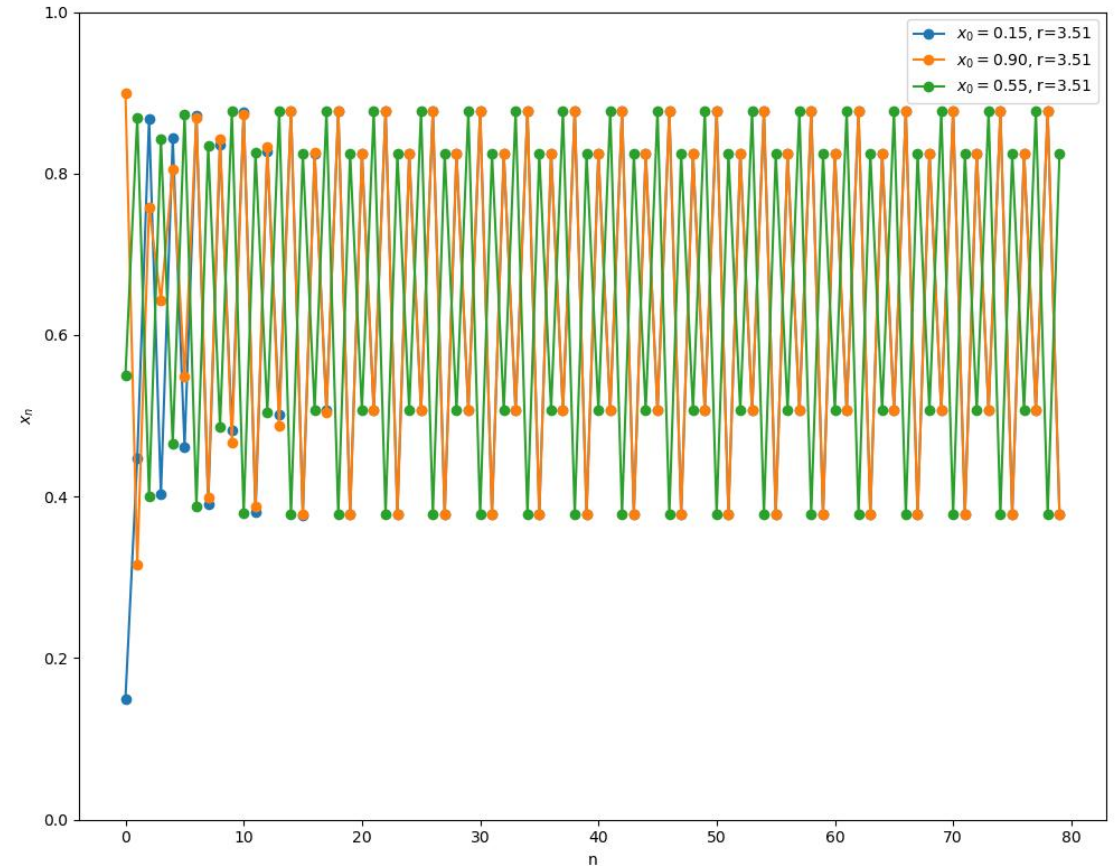
Behavior of the logistic map: beyond 3

- $x_{n+1} = rx_n(1 - x_n)$
- Let's consider **r=3.2** this time
- Regardless of the initial value **the population is... attracted to...?**
- It always oscillates **between two values**, no matter the initial value
- This behavior only happens in the interval **[3, 1+√6]** ($1 + \sqrt{6} \approx 3.449489 \dots$)



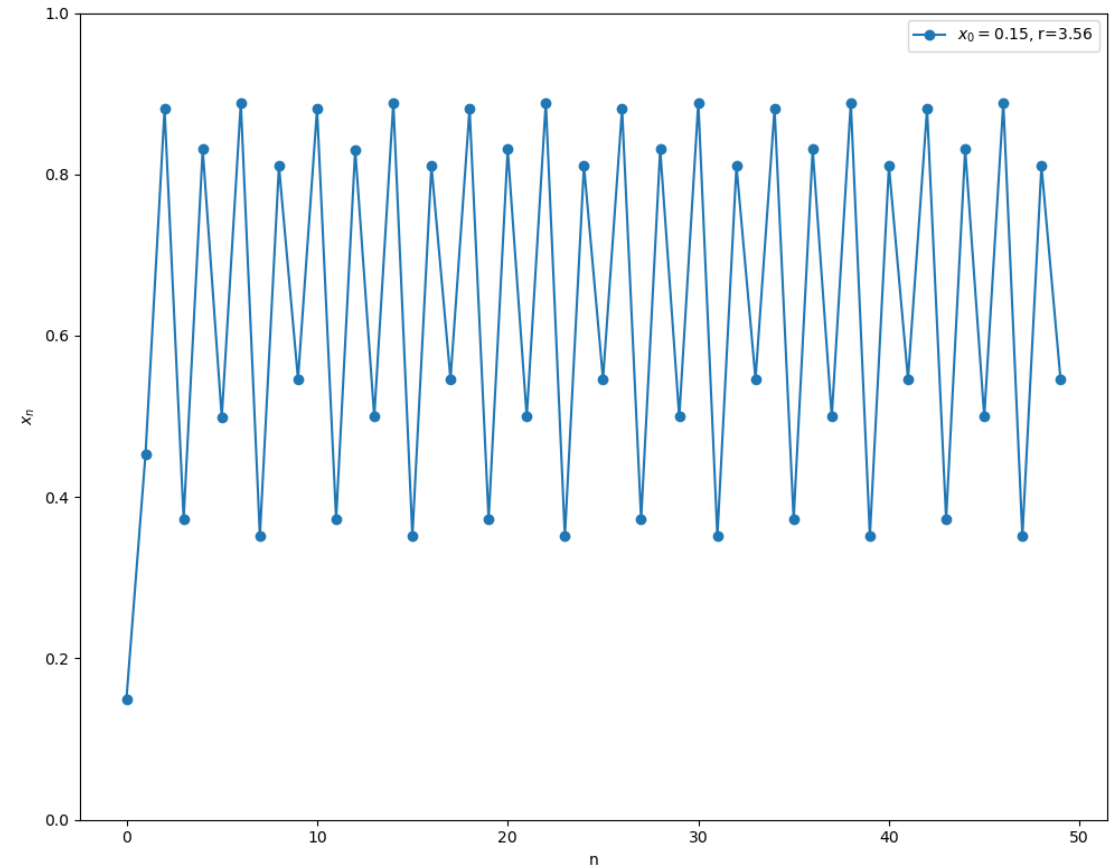
Behavior of the logistic map: $r=3.51$

- $x_{n+1} = rx_n(1 - x_n)$
- Let's consider $r=3.51$
- This time it **oscillates** between **four** values, regardless of the initial value
- This behavior only happens in the interval **[3.44949, 3.54409]**



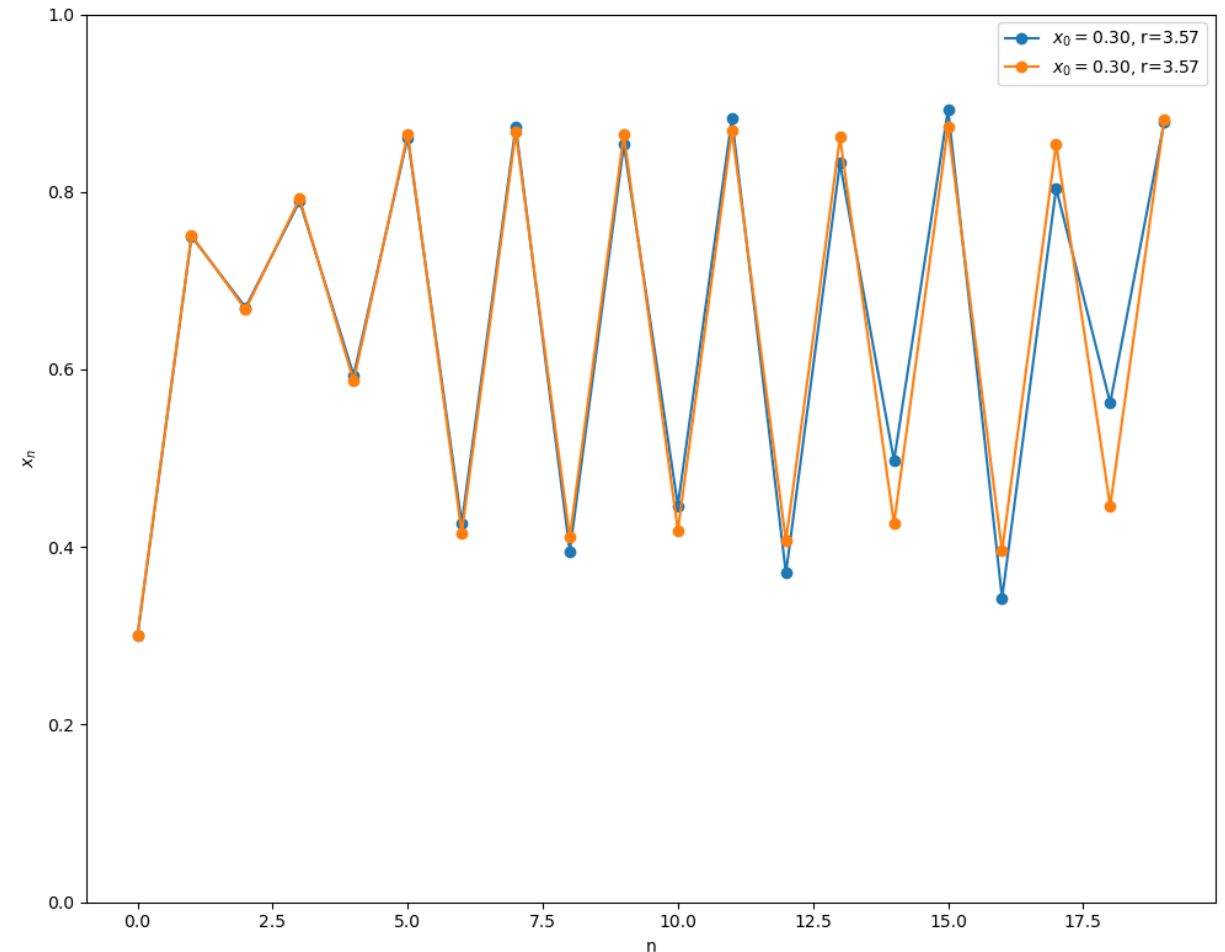
Behavior of the logistic map: $r=3.555$

- $x_{n+1} = rx_n(1 - x_n)$
- Let's consider $r=3.555$
- This time it oscillates **between eight values**, regardless of the initial value
- Further values will lead to a **doubling** of oscillations: 16, 32, etc.
- This behavior only happens in the interval **[3.5504, 3.56995]**



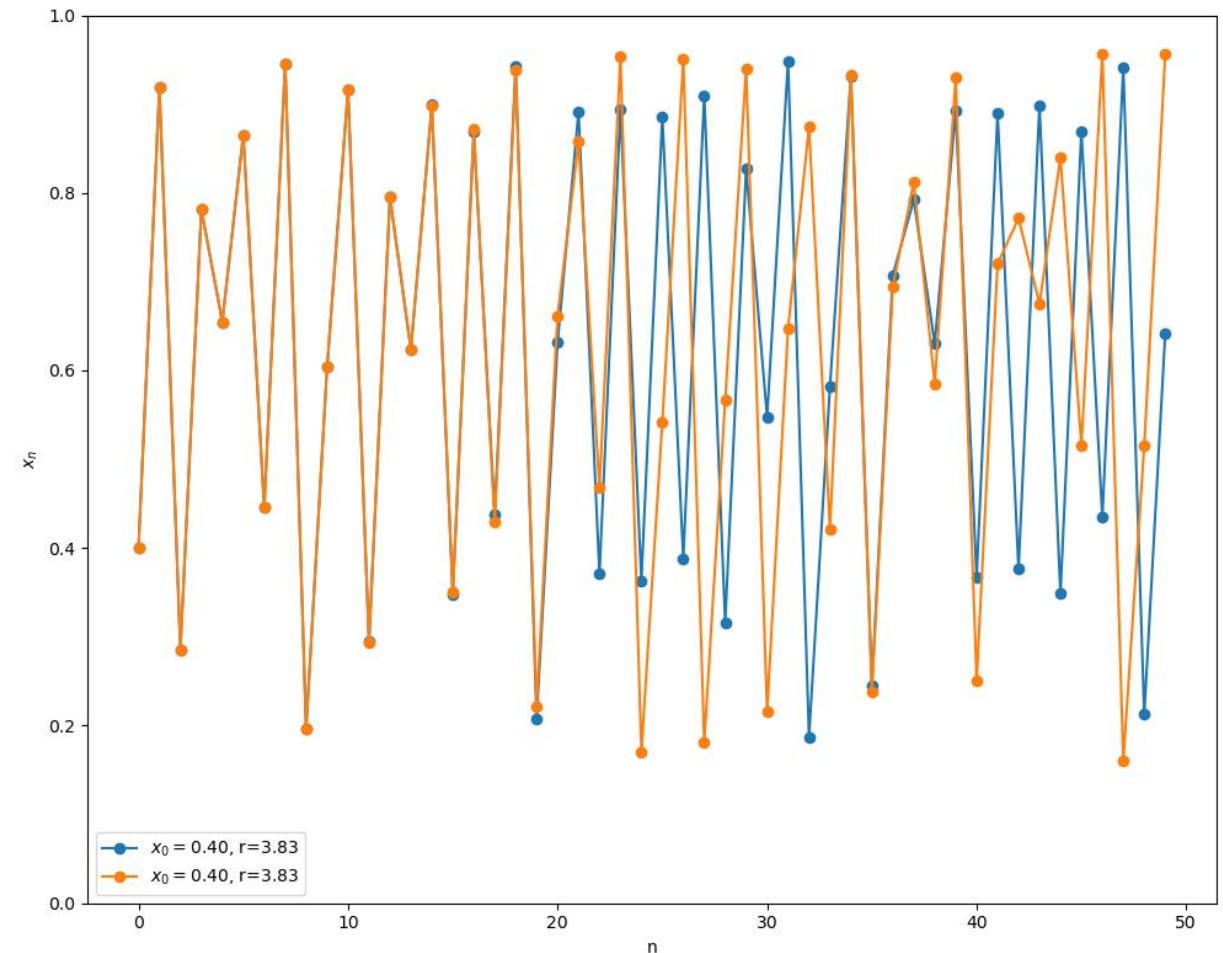
The verge of Chaos: $r \approx 3.56995$

- $x_{n+1} = rx_n(1 - x_n)$
- From this value on, **something weird** begins to happen
- **Similar initial conditions do not lead to similar trajectories**
- **A small variation leads to dramatically different results**
- A dynamical system in which **a small change** in the initial conditions leads to a **big change** in the final state is called **chaotic**



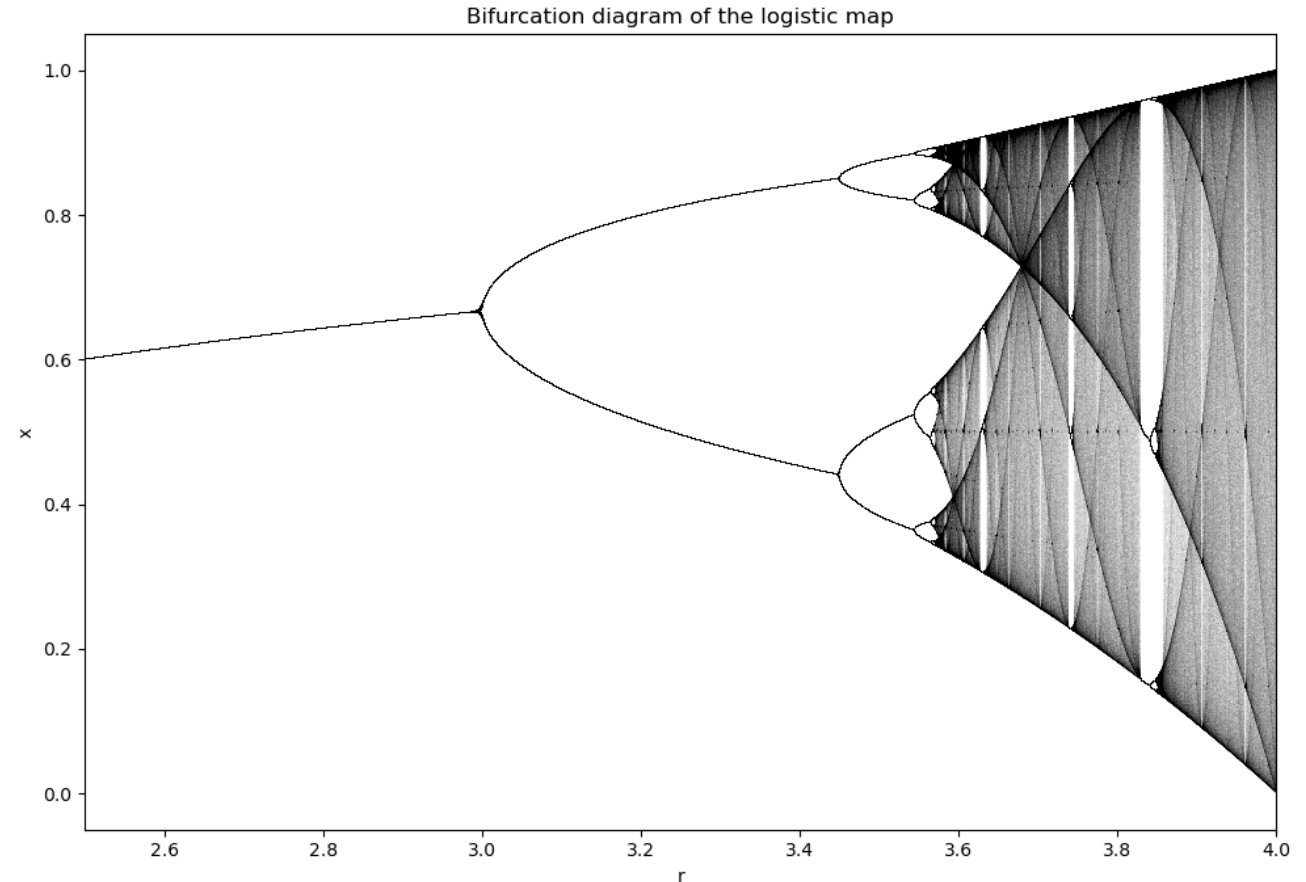
Chaos: $r=3.82843$

- $x_{n+1} = rx_n(1 - x_n)$
- Here the initial conditions I used were 0.4 and $0.4+1\mathbf{e-6}$ (!)
- **The two dynamics diverge completely** within a few iterations
- It is basically **impossible** to determine whether there are any **stable attractors**
- Remember: the system is **deterministic**, no randomness whatsoever



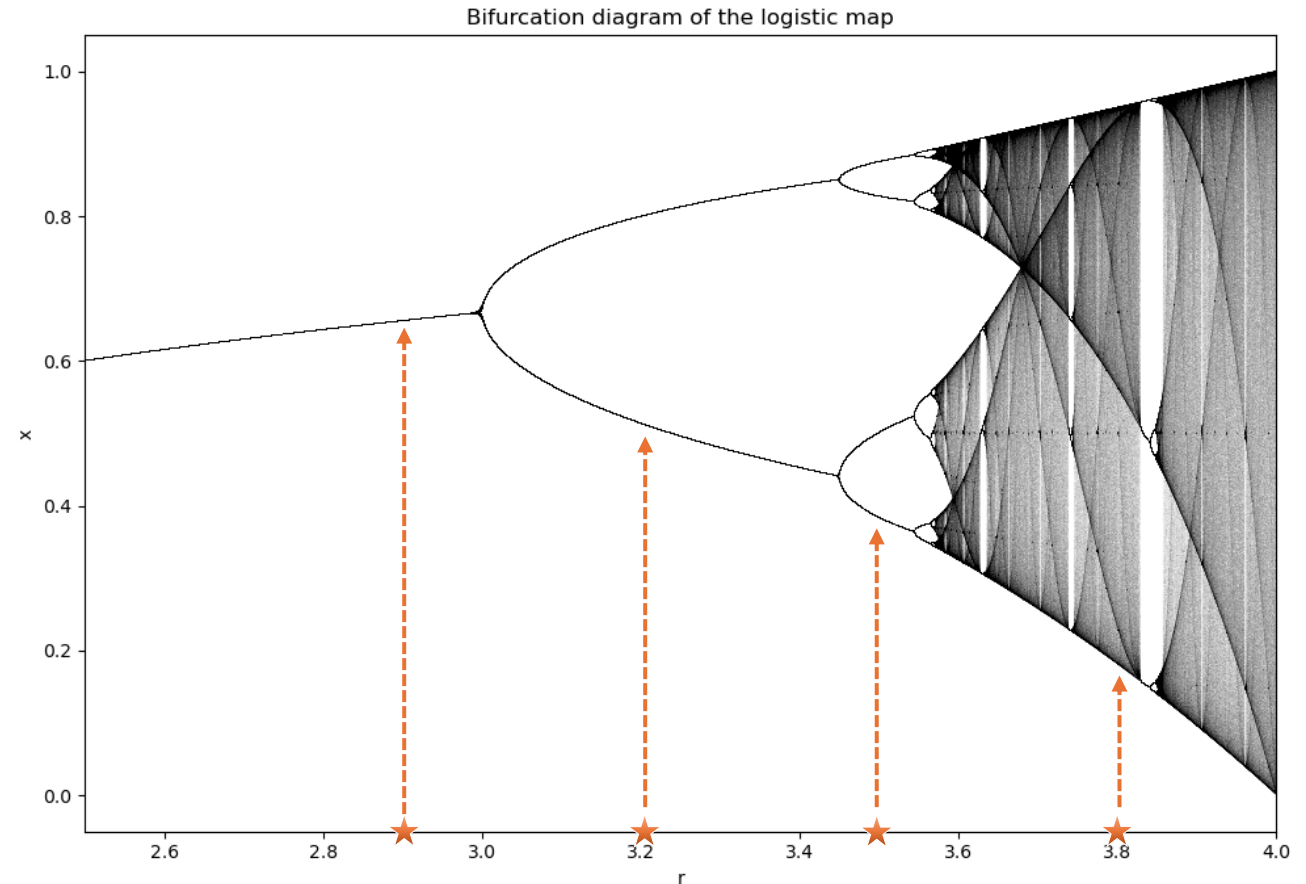
Bifurcation diagram

- The diagram shows how the long-term **population of a species changes** as the **growth rate parameter is varied**
- For **low values** of r , the population stabilizes around a **single value**
- As the growth rate increases, the population starts **oscillating between two values**, then four, and eventually enters the **chaotic regime**



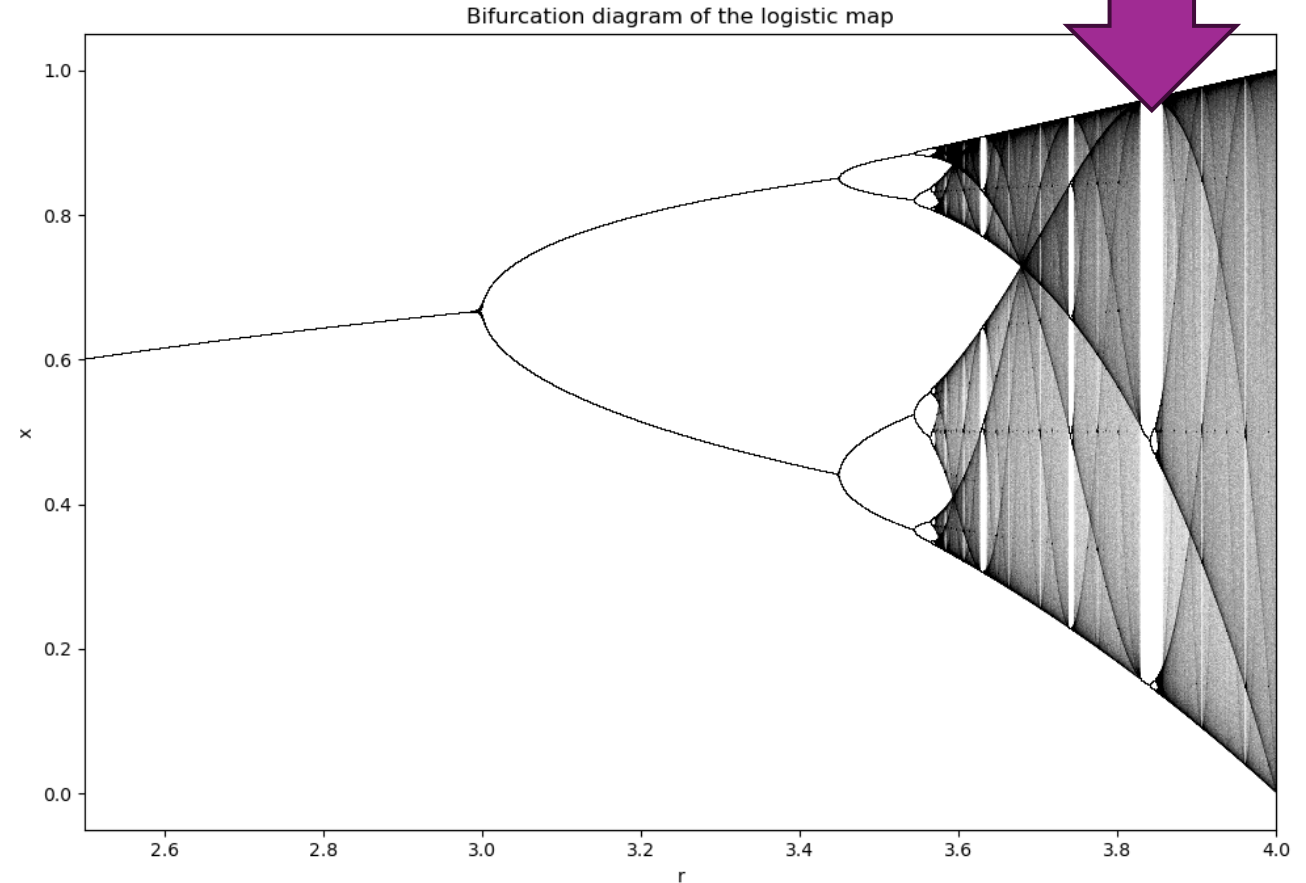
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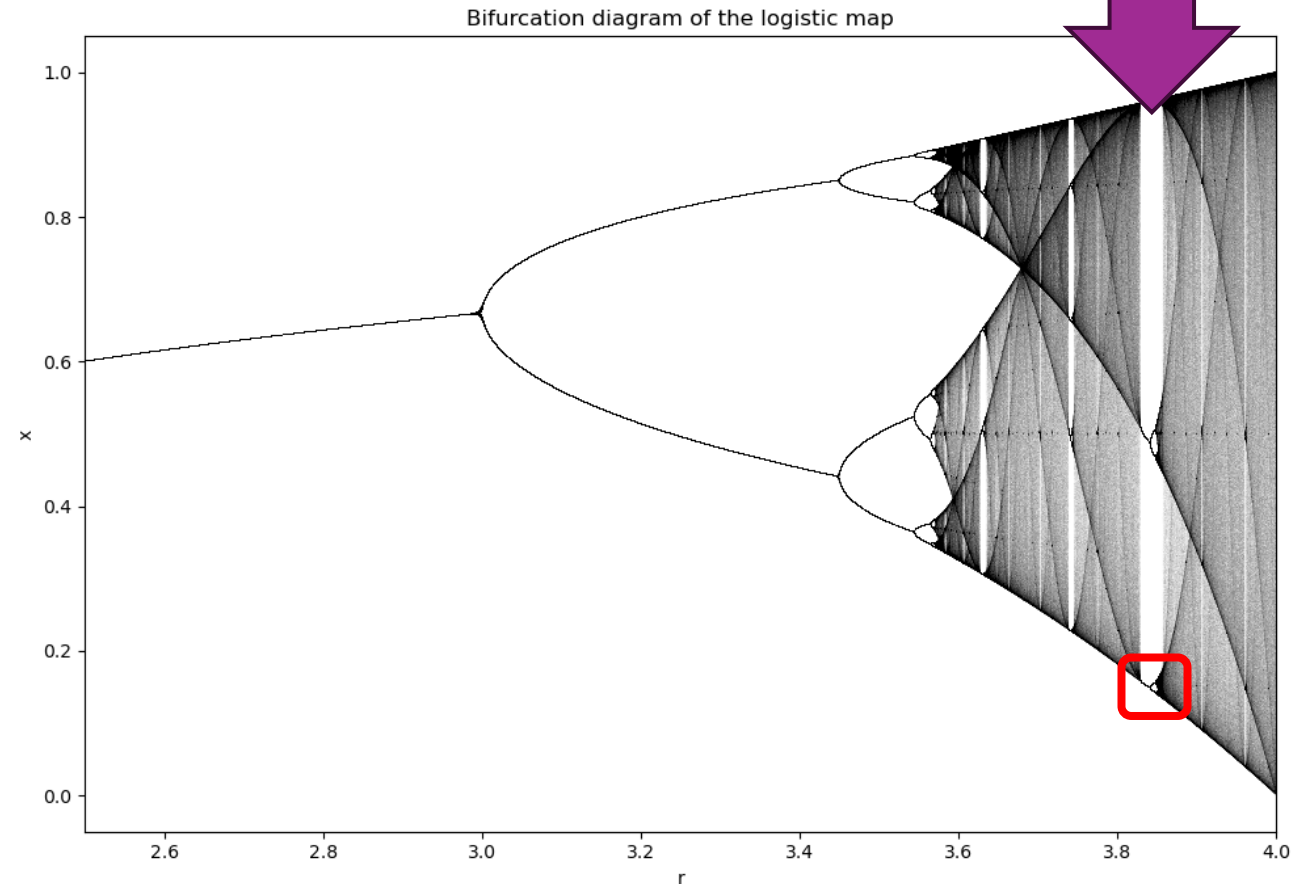
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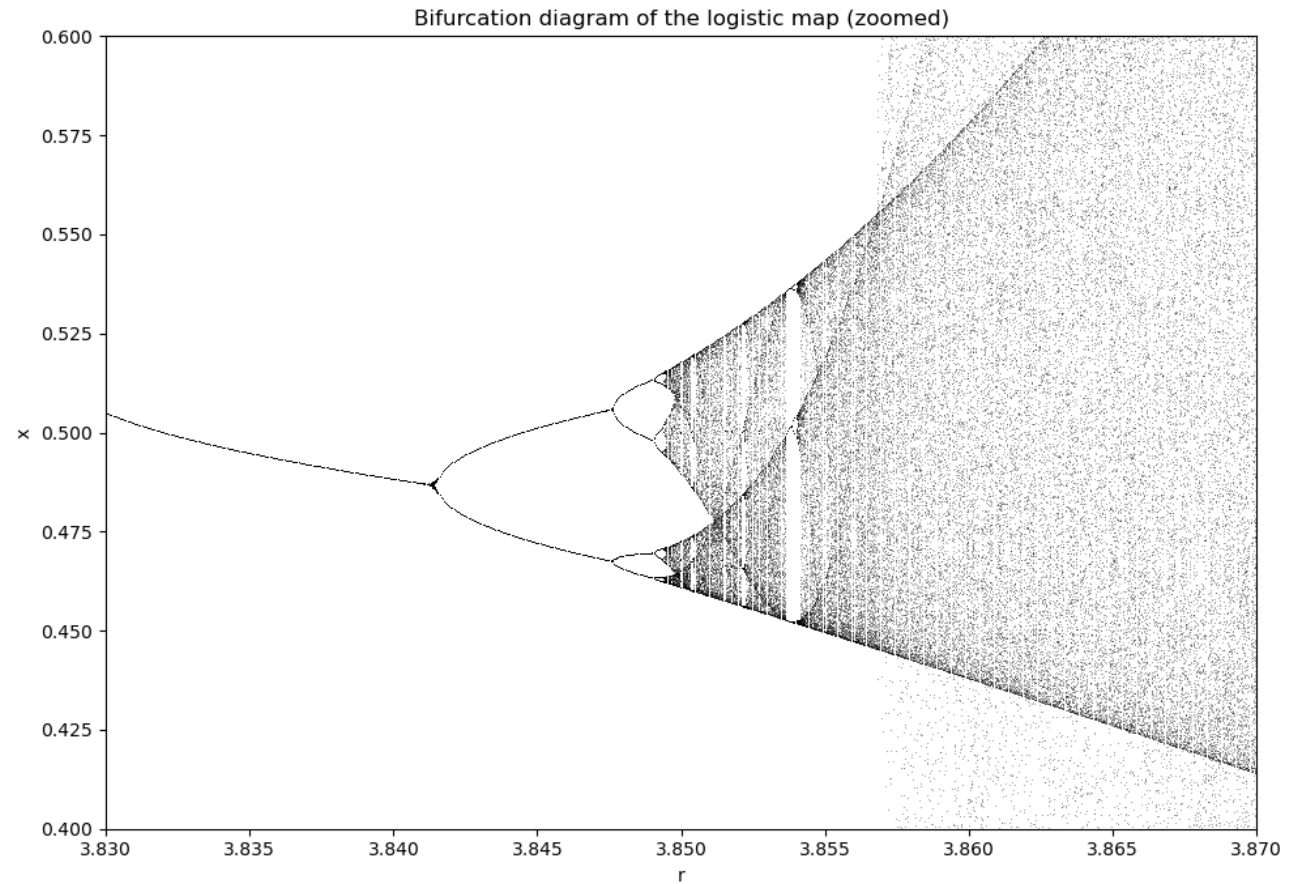
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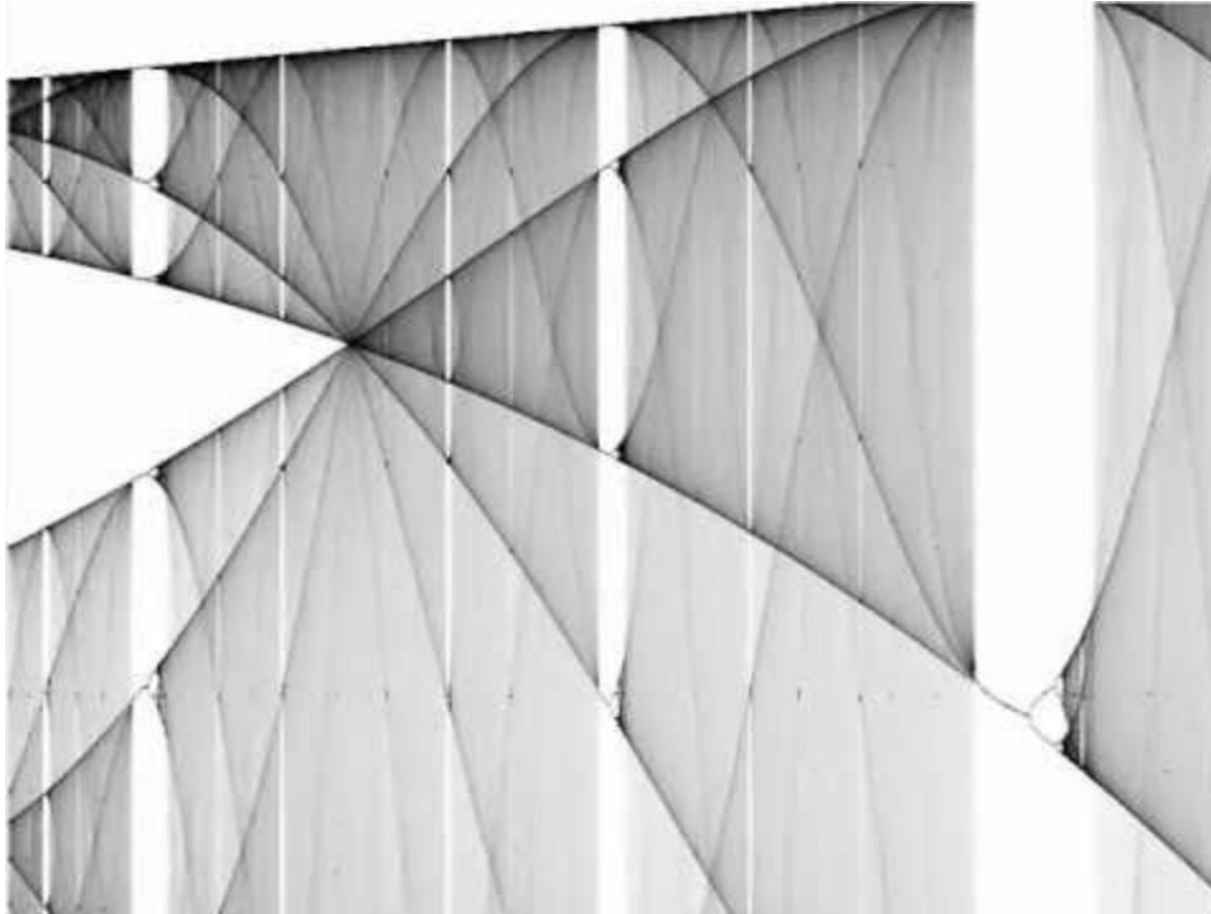


Self-similarity of bifurcations

- **Self-similarity**: an object is **similar to a part** of itself
- We find the same bifurcation diagram over and over again
- Self-similarity is a typical property of **fractals**

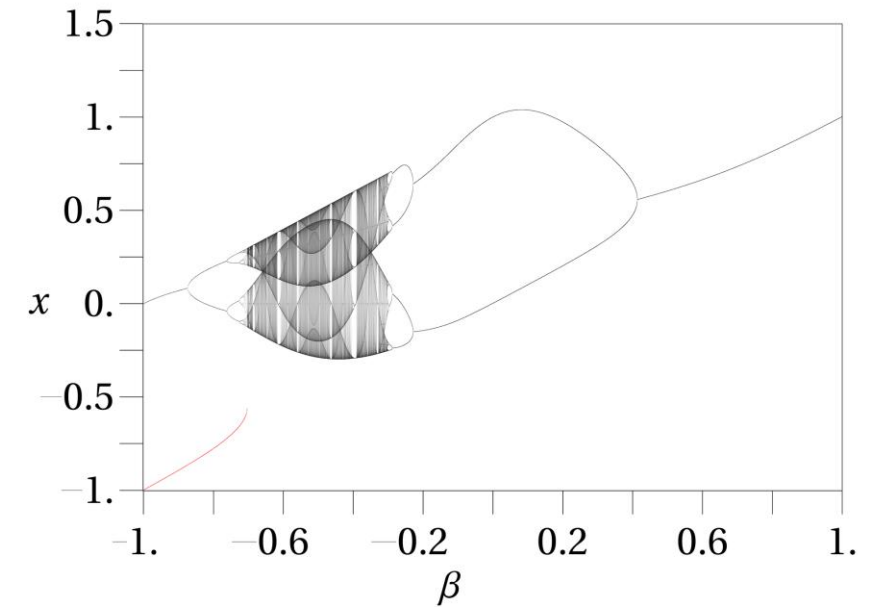


Bifurcation diagram magnified 1M times

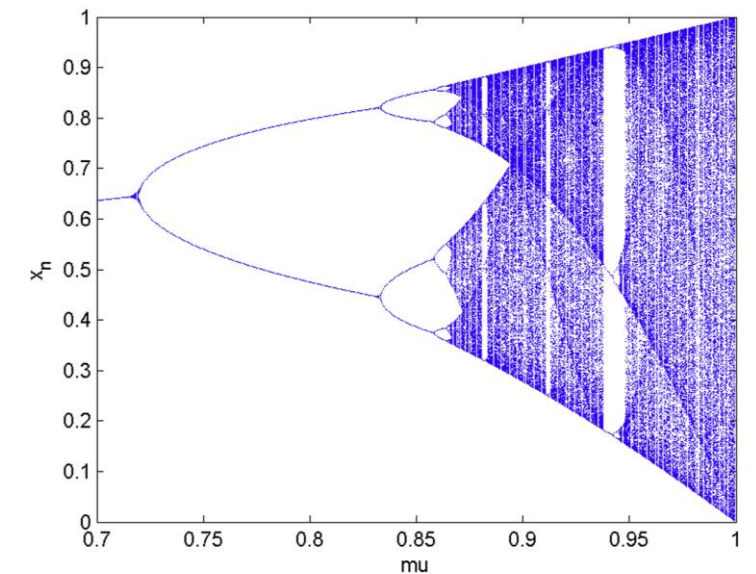


What else bifurcates?

- Gauss map: $x_{n+1} = \exp(-\alpha x_n^2) + \beta$



- Sine map: $x_{n+1} = \mu \cdot \sin(\pi x_n)$



Let's consider a further map

$$x_{n+1} = x_n^2 + c$$

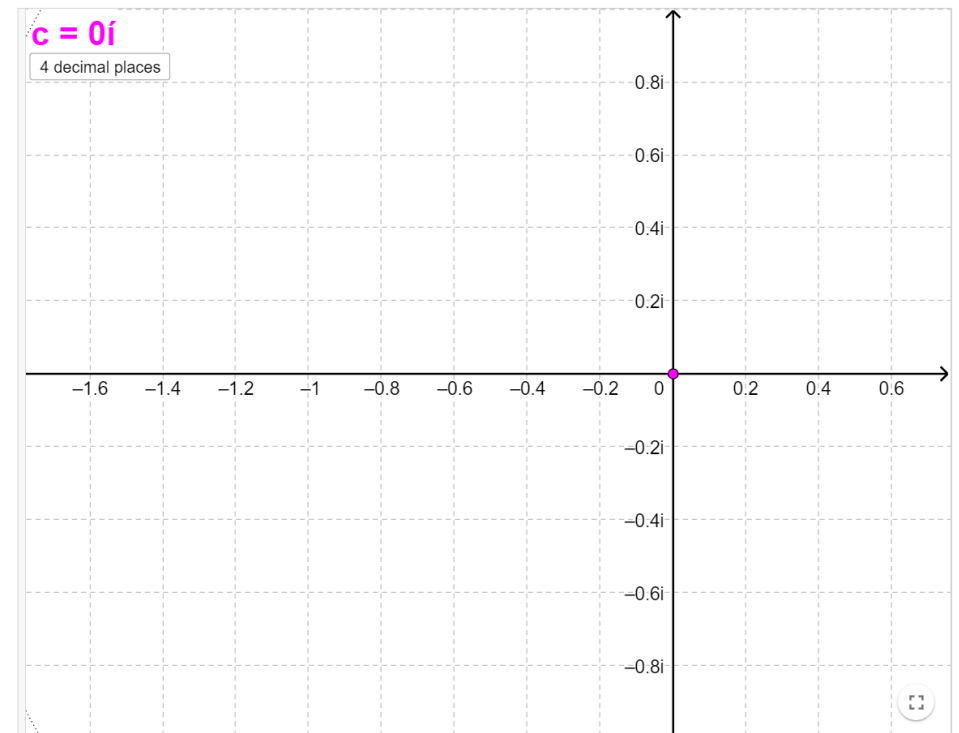
- Differently from all previous examples, this time we consider $x \in \mathbb{C}$
- Remember that the square of a complex number c is the product of that number against its complex conjugate c^* and $i^2 = -1$
(that is, $c = a + ib, c^* = a - ib$)
- Thus, $c^2 = c \cdot c^* = a^2 + b^2$

The orbits of the $x_{n+1} = x_n^2 + c$ map

- We might wonder whether the **sequence converges** to some stable point or **diverges**
- Well, if $x_0 = 0 + 0i$, the orbit does not move

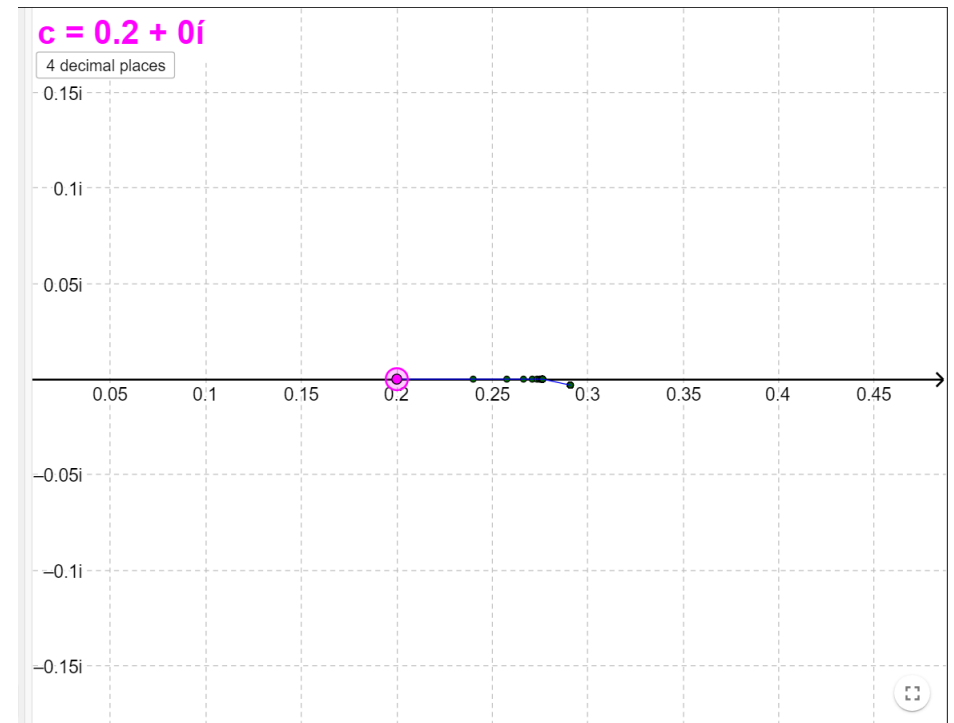
<https://www.geogebra.org/m/mfewjrek>

This is the
complex plane



The orbits of the $x_{n+1} = x_n^2 + c$ map

- We might wonder whether the **sequence converges** to some stable point or **diverges**
- Well, if $x_0 = 0.2 + 0i$, the orbit **converges** after a few steps



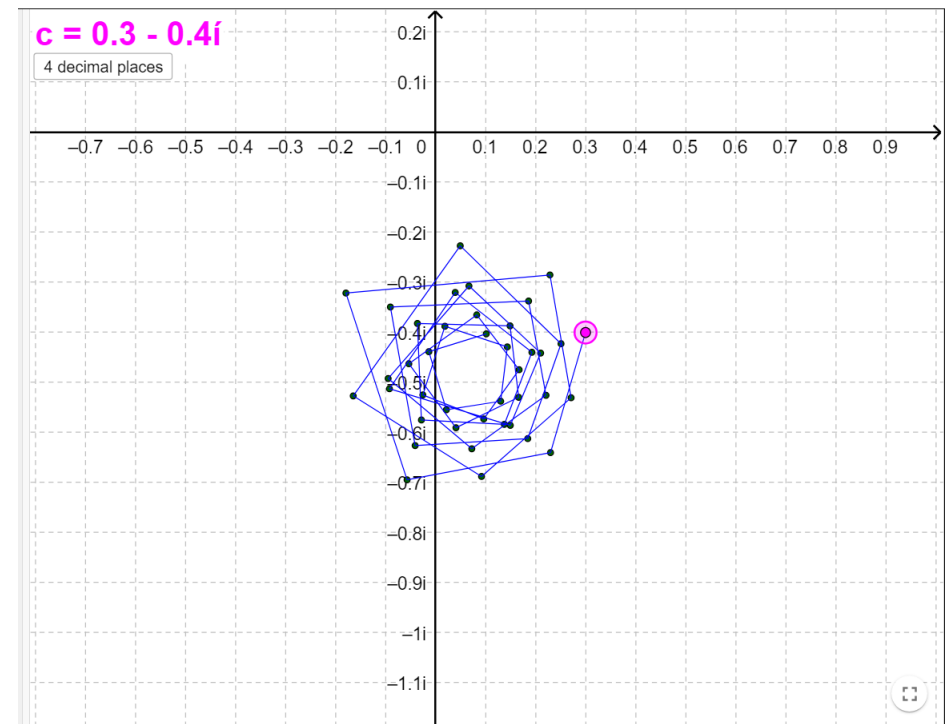
The orbits of the $x_{n+1} = x_n^2 + c$ map

- We might wonder whether the **sequence converges** to some stable point or **diverges**
- Well, if $x_0 = 0.3 + 0i$, the orbit diverges to **infinity**
- Let's try to abandon the real axis



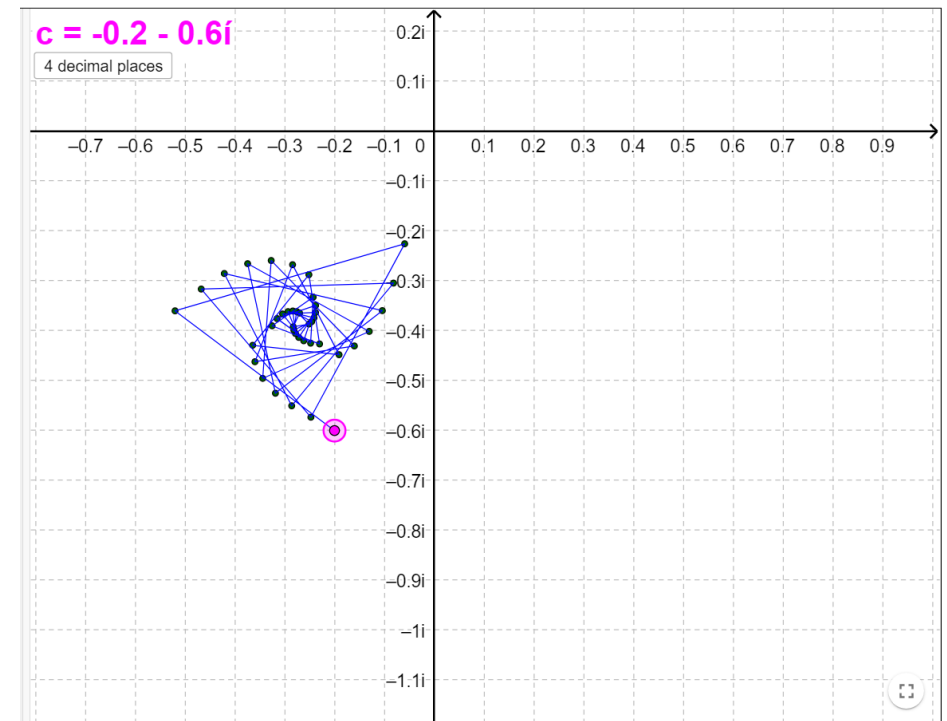
The orbits of the $x_{n+1} = x_n^2 + c$ map

- We might wonder whether the **sequence converges** to some stable point or **diverges**
- If $x_0 = 0.1 - 0.4i$, the sequence converges with a «**circular**» **orbit**



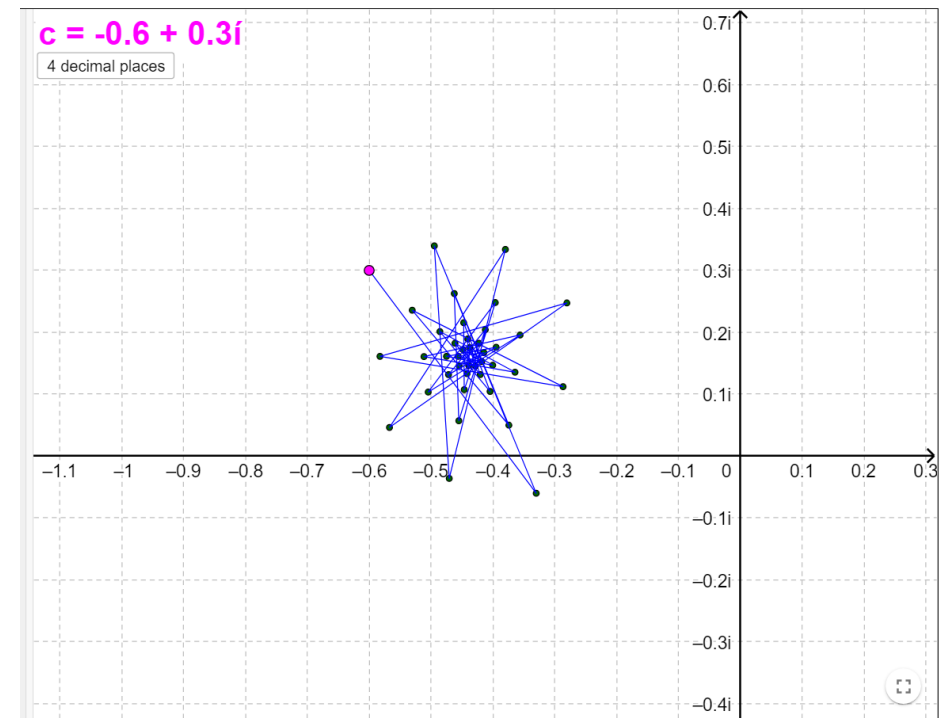
The orbits of the $x_{n+1} = x_n^2 + c$ map

- We might wonder whether the **sequence converges** to some stable point or **diverges**
- If $x_0 = -0.2 - 0.6i$, the sequence converges with a... **peculiar orbit**



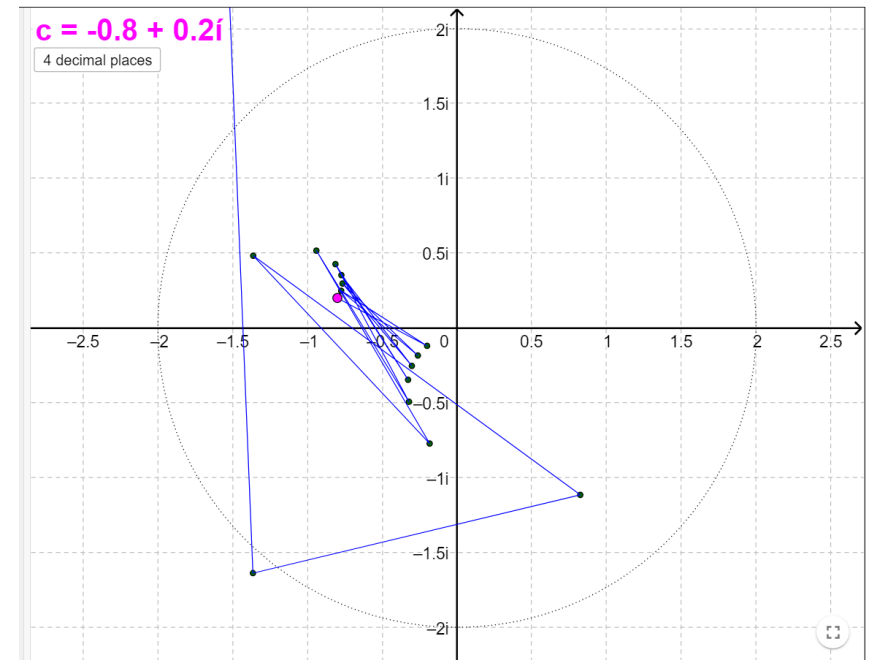
The orbits of the $x_{n+1} = x_n^2 + c$ map

- We might wonder whether the **sequence converges** to some stable point or **diverges**
- If $x_0 = -0.6 + 0.3i$, the sequence converges with a even more star-shaped **orbit**



The orbits of the $x_{n+1} = x_n^2 + c$ map

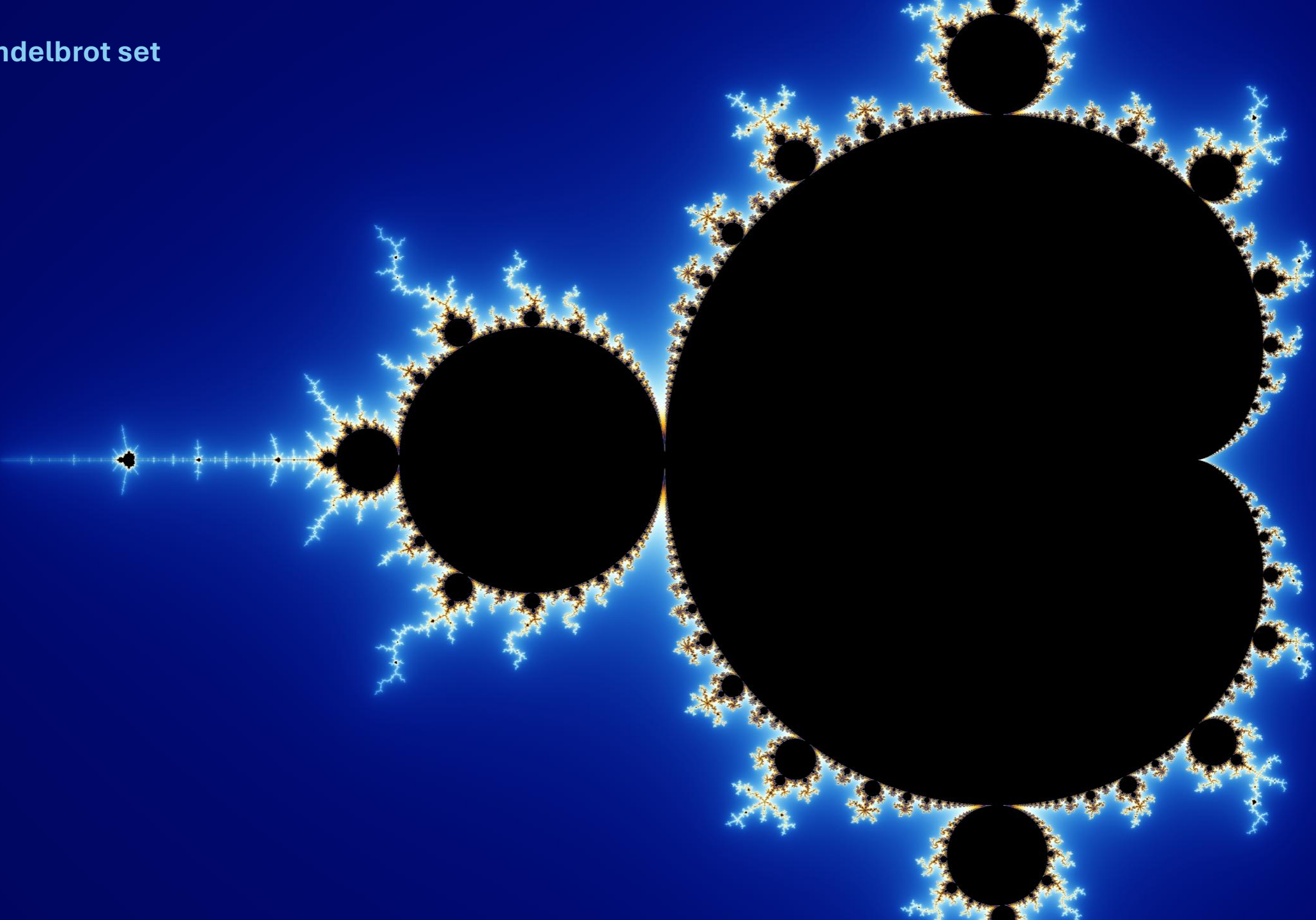
- We might wonder whether the **sequence converges** to some stable point or **diverges**
- If $x_0 = -0.8 + 0.2i$ the **orbit diverges**
- Divergence is certain when x goes outside the radius 2 circle (dotted)



Let's draw a diagram

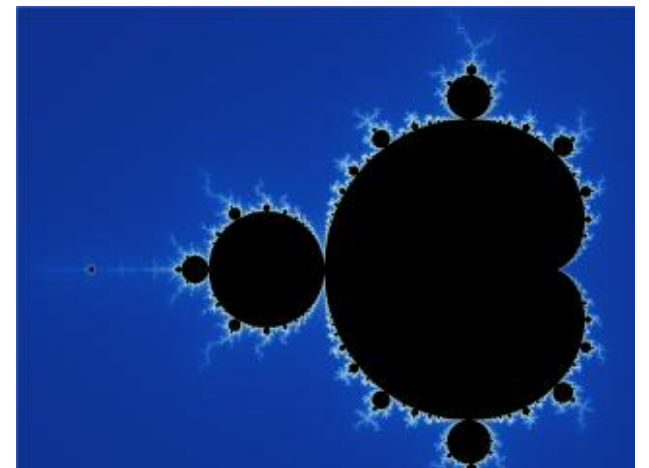
- Some points lead to **stable orbits**, some other points lead to an **unstable** (divergent) **orbit**
- Let's draw **points** leading to **stable orbits in black**
- Let's use **color to draw points whose orbit diverges**, and the color **denotes how fast** an orbit to diverges

Mandelbrot set



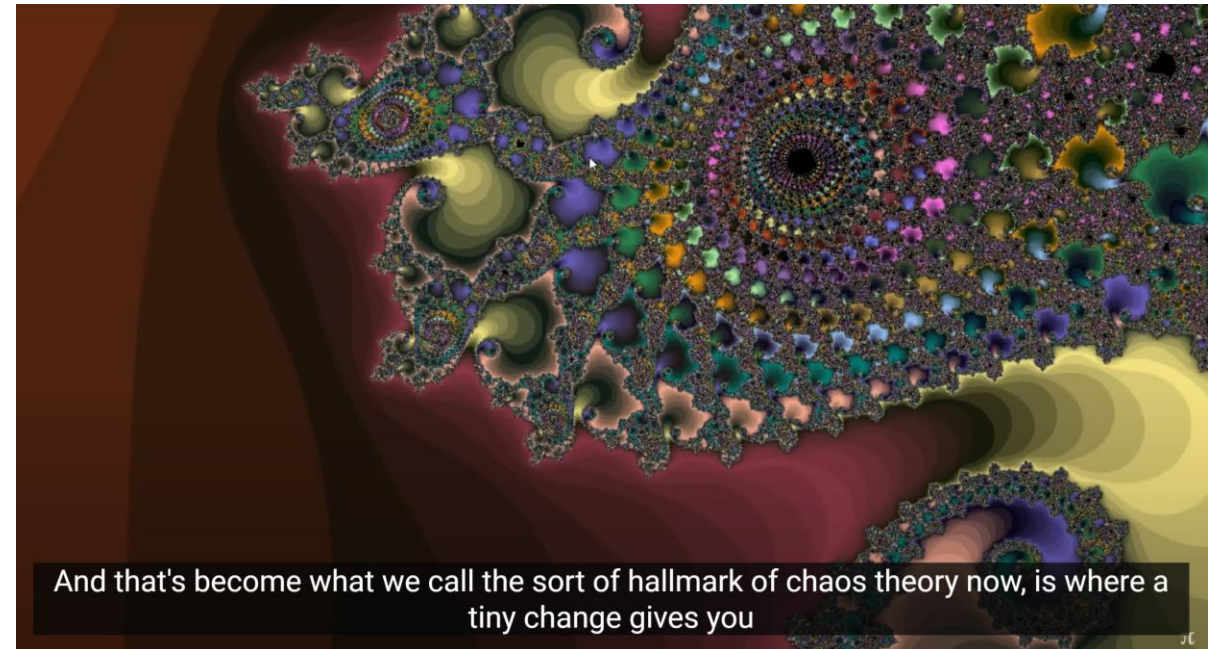
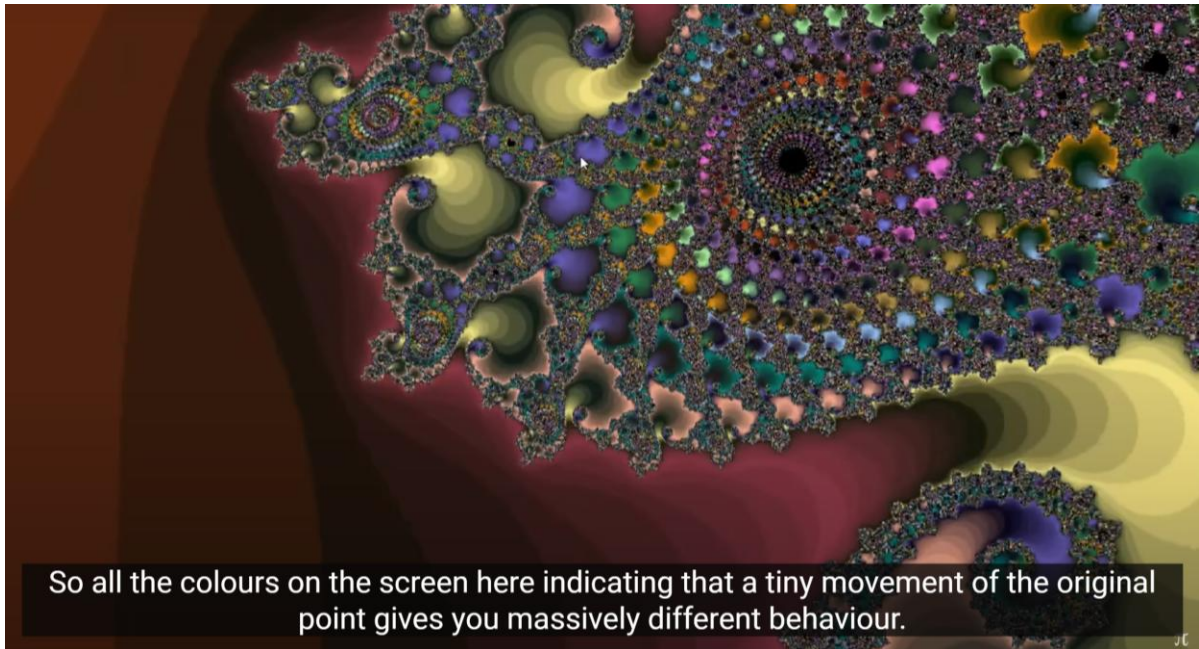
Benoît Mandelbrot

- Mathematician, well known for **fractal geometry**
- His goal was to investigate the concepts of **roughness and self-similarity** in nature
- He clearly discovered Mandelbrot's set
- Mandelbrot's set is **self-similar**



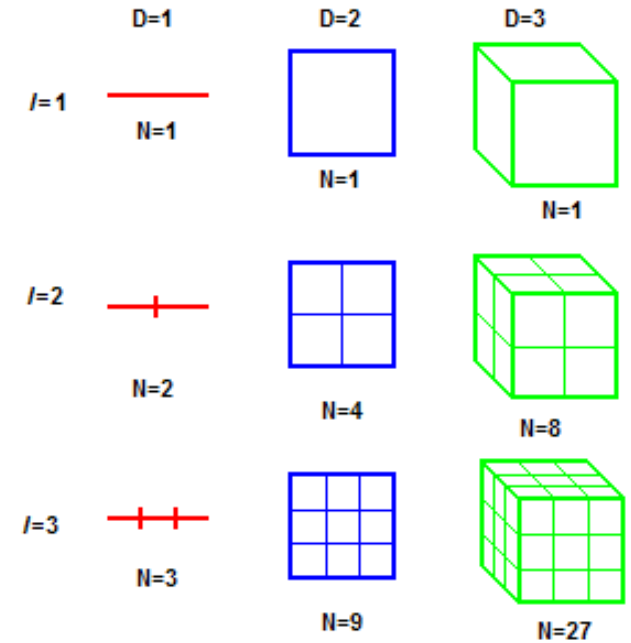
Mandelbrot's set is chaotic

- Remember that color of the point denotes the **stability of the orbit**
- A **tiny change** of the initial point can lead to **radically different** orbits → that's the very definition of a **chaotic system**
- www.youtube.com/watch?v=FFftmWSzgmK



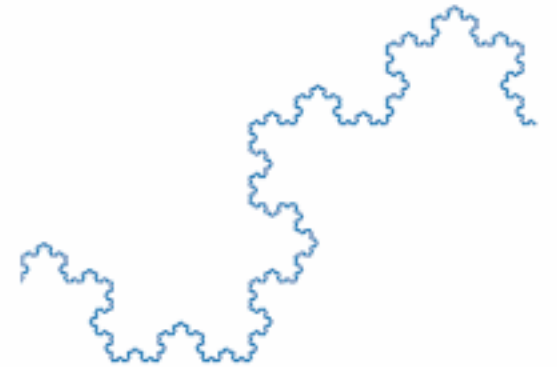
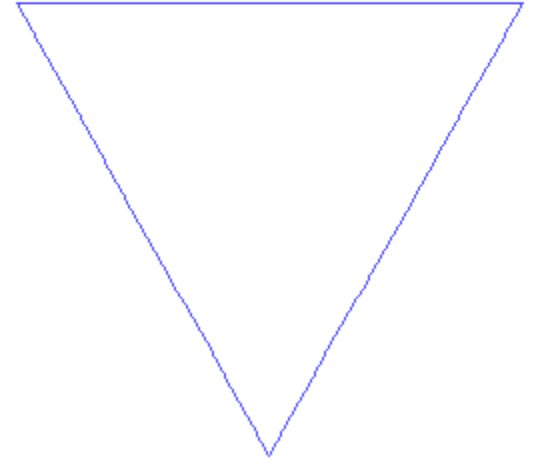
Fractals==self-similarity?

- Not exactly that...
- **Fractal Dimension (FD)**: the ratio of the **change in detail** to the **change of scale**
 - Imagine to count the **unitary segments** on the **real line**
 - If we «zoom out» the number of segments increases linearly
 - The same happens with squares in 2D and cubes in 3D
- If ε is the scaling factor and N the number of «counting units», then D corresponds to the fractal dimension according to the formula: $N = \varepsilon^{-D}$
 - That is $D = -\log_{\varepsilon} N$
 - For instance, in this last example, $D = \log_3 27 = 3$ (nothing new...)



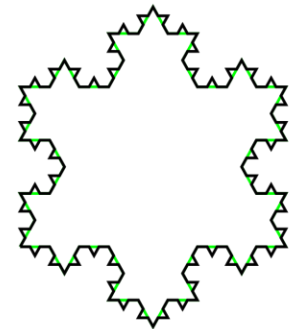
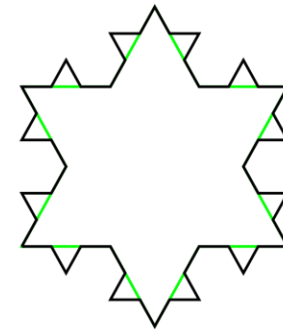
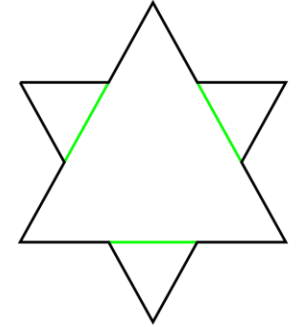
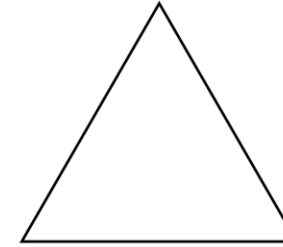
FD of the Koch snowflake fractal

- The **Koch snowflake** is an iterative shape created by a **fractal curve**
 1. It begins with an **equilateral triangle**
 2. Each side is **subdivided into three equal segments**
 3. The second segment introduces a further, **smaller equilateral triangle**
 4. Repeat from #2



FD of the Koch snowflake fractal

- Initially this curve has length 3
- Measured with a unit with scale $1/3$ it has length 4 instead of the expected 3 (!)
- The fractal length is $-\log_{\varepsilon} N = \log_{\frac{1}{3}} 4 = 1.26\dots$
- The more detail we add, the more the perimeter increases ($\rightarrow$ infinite perimeter! Wait, what?!)
- Interestingly, the area is finite (Wait, what2?!)

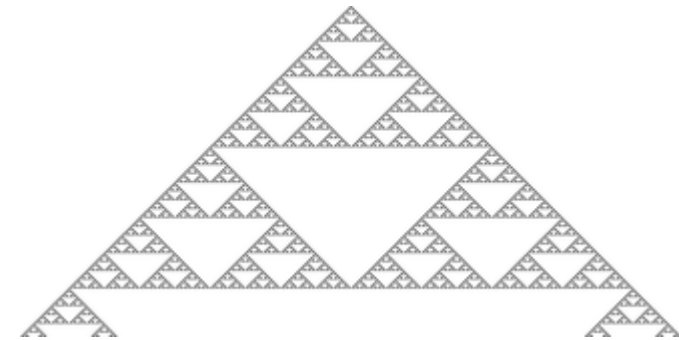


Wacław Sierpinski

- One last fractal: **Sierpiński's triangle**

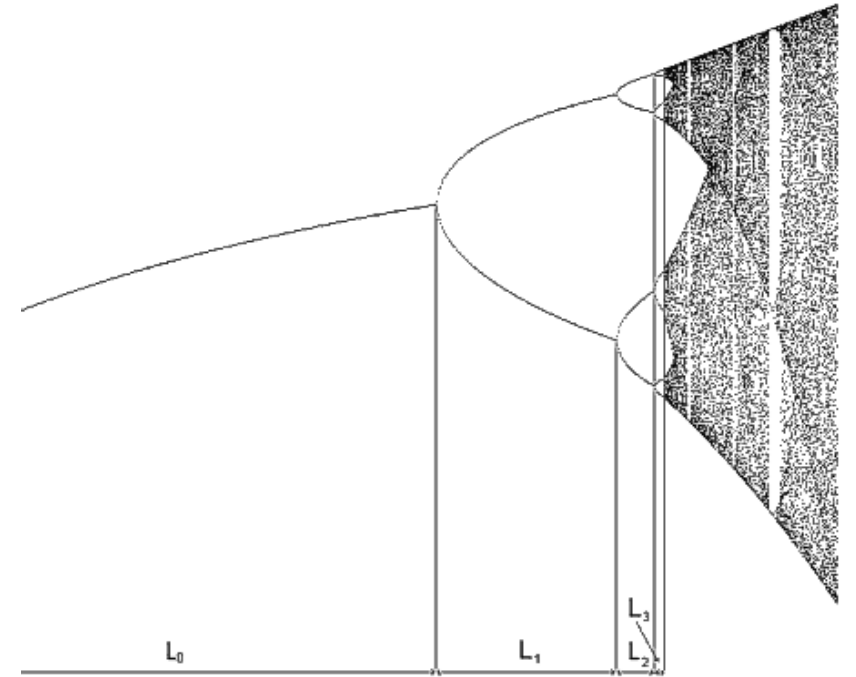


- Fractal dimension = 1.585
- At each step, the area becomes $\frac{3}{4}$ of the original
- Thus, after n iterations, it is equal to $\left(\frac{3}{4}\right)^n$ and the limit for $n \rightarrow \infty \dots$



The Feigenbaum constant

- Let's get back to the logistic map. In 1975, the mathematician Mitchell Feigenbaum observed that the **ratio between period-doubling bifurcations** ($\frac{L_i}{L_{i+1}}$) tends to a constant δ
- Assume $x_{n+1} = \mu f(x_n)$, we can define as μ_i the value that delimits the i-th bifurcation point
- Then $\delta = \lim_{n \rightarrow \infty} \frac{\mu_{n+1} - \mu_n}{\mu_{n+2} - \mu_{n+1}} = 4,66920160910299067185320382 \dots$



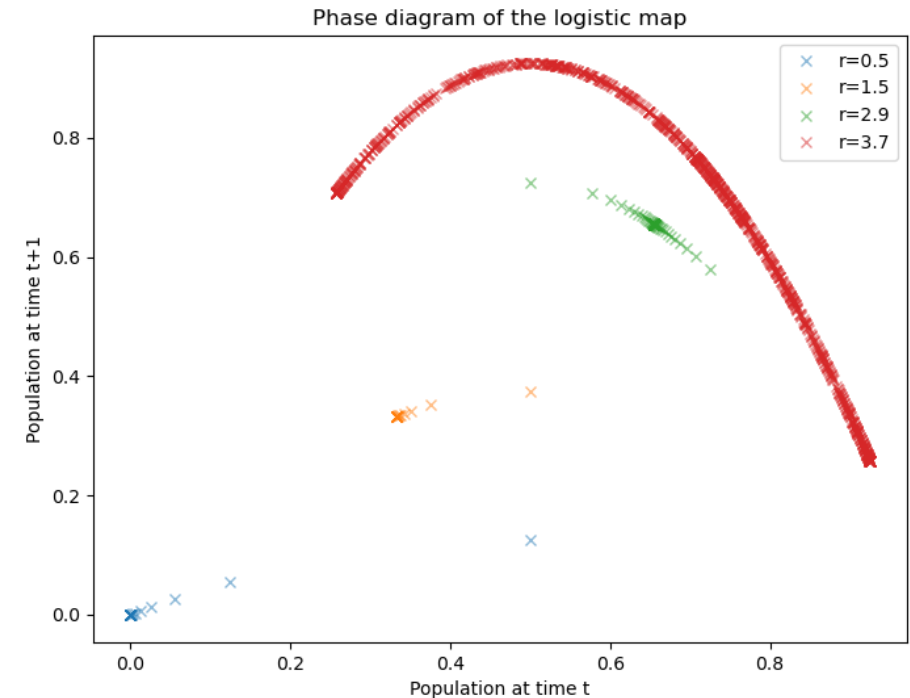
The Feigenbaum constant seems universal

- Besides the logistic map, the Feigenbaum constant **arises with *many other maps***
 - $x_{n+1} = a - x_n^2$
 - $x_{n+1} = r \cdot \sin(\pi x_n)$
 - $x_{n+1} = r \cdot \min(x_n, 1 - x_n)$
 - $x_{n+1} = r \cdot x_n(1 - x_n^2)$
 - ...
- It even arises with the **Mandelbrot set**, as the limit ratio between the diameters of successive circles on the real axis in the complex plane



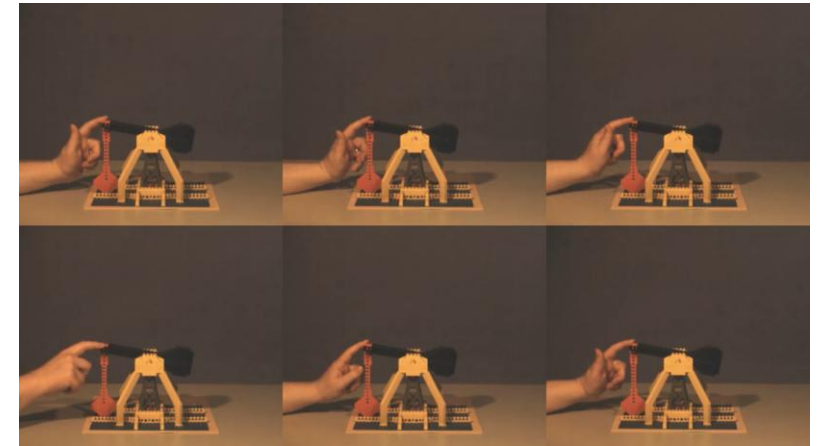
Phase diagrams

- Let's try to plot the population at time t and $t + 1$ on the x and y axis
- What we get is the so-called «**phase diagram**», useful to investigate the basins of attraction
- The phase diagram clearly shows that small values of r lead to some attractors, $r = 2.9$ oscillates around multiple values, and $r = 3.7$ is... all over the place!

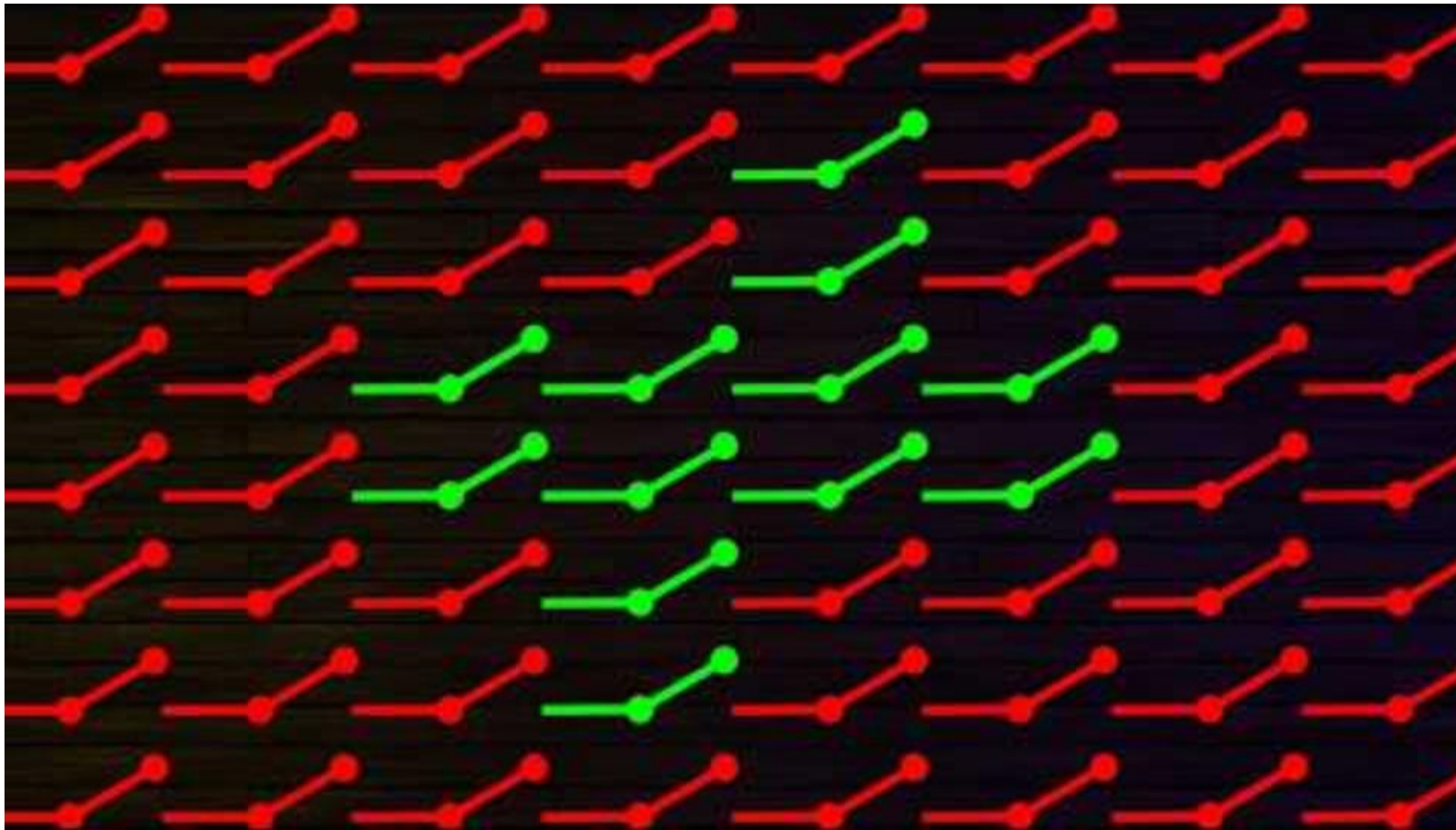


The «butterfly effect»

- **Chaotic** systems are **extremely sensitive to initial conditions**
- This is the origin of the infamous **butterfly effect**
- Thus a **small modification of the initial conditions** (e.g., the flap of a butterfly) can have a **huge impact to the outcome** of the simulation (e.g., a change in the weather, possibly a storm)
- We never know with absolute certainty the initial conditions of a system → chaotic systems are tricky to simulate and investigate
- One real-world example: [double pendulum](#)

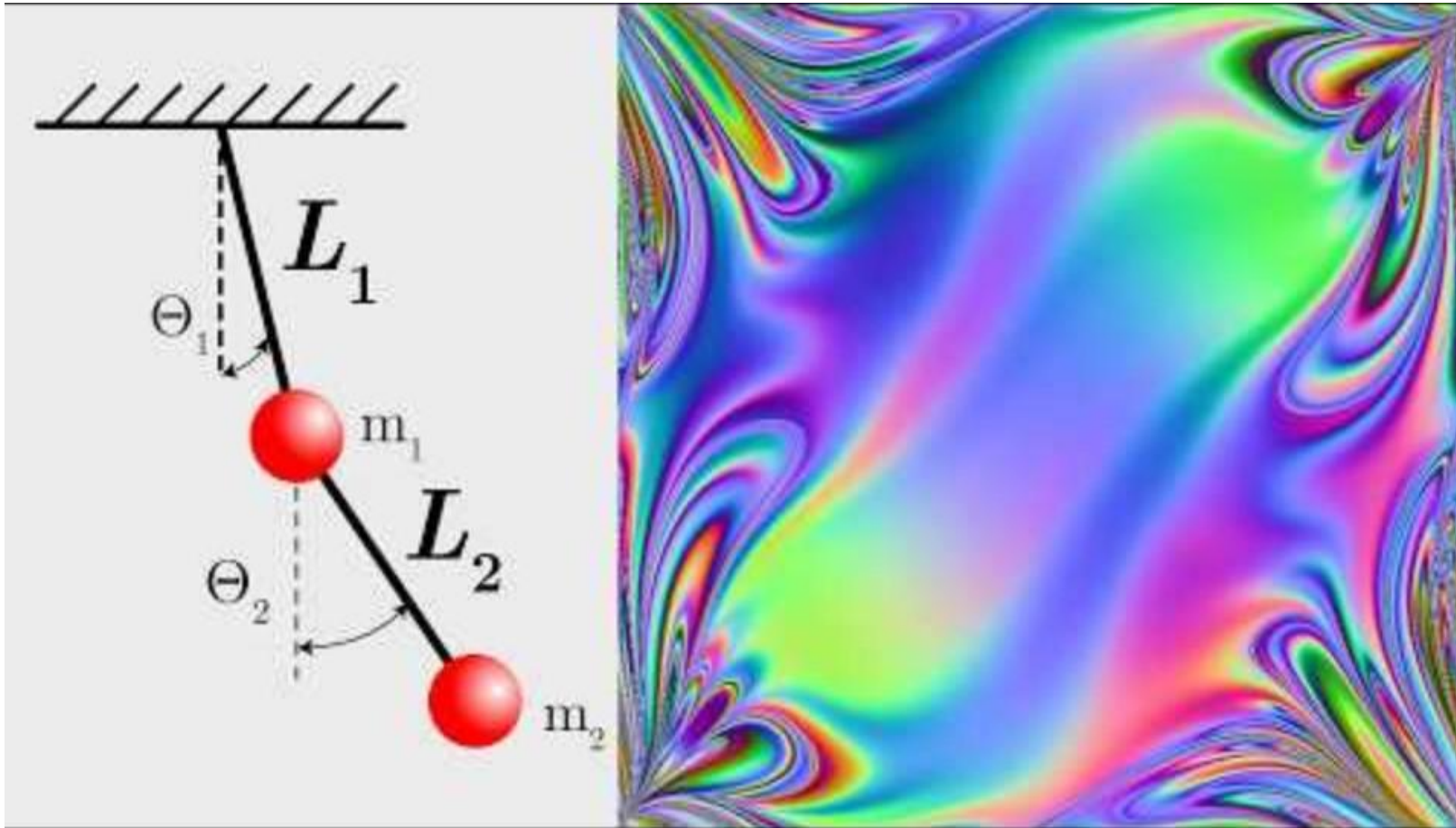


Island of stability: double pendulum edition



<https://www.youtube.com/watch?v=OIHyN7TzY6A>

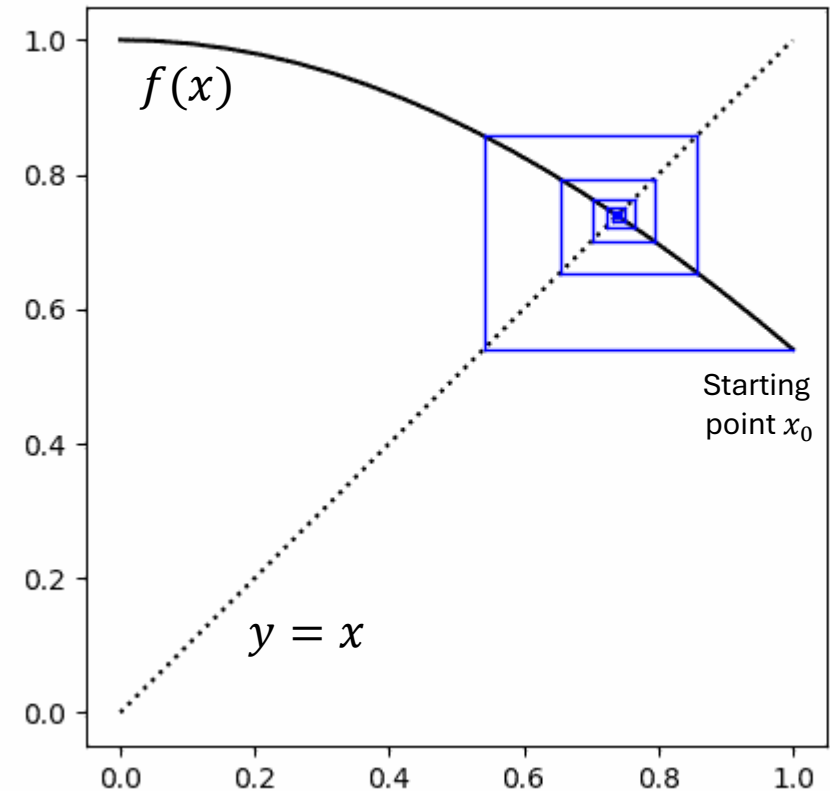
Fractals of stability islands



https://www.youtube.com/watch?v=K5dOm_WWaNY

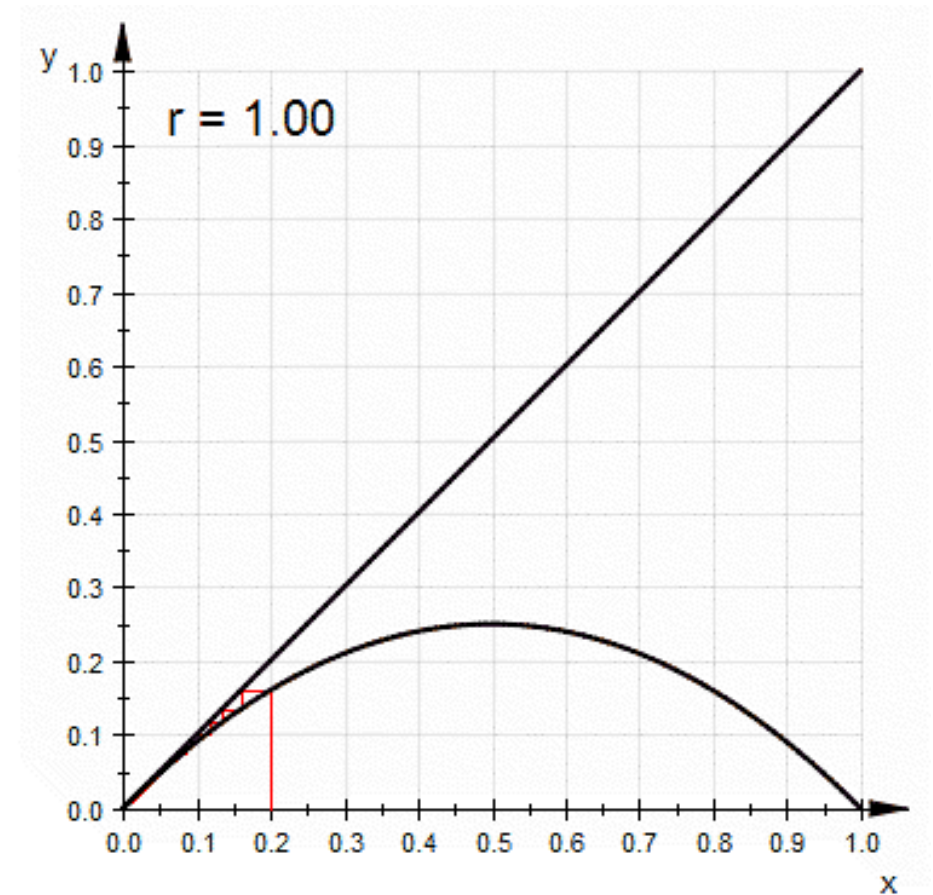
Cobweb plots

- Aka Verhulst diagram, it is used to analyse the behavior of **dynamical systems**
- The diagram shows the long-term fate of an **iterated function** $f: \mathbb{R} \rightarrow \mathbb{R}$
 - E.g., $f(x) = rx(1 - x)$
- How to create a Cobweb plot:
 1. Draw the line $y = x$ and plot the function $f(x)$
 2. Pick a value x_0 and set $x = x_0$
 3. Identify the point $(x, f(x))$
 4. Draw the horizontal segment between $(x, f(x))$ and $(f(x), f(x))$
 5. Draw the vertical segment between $(f(x), f(x))$ and $(f(x), f(f(x)))$
 6. Iterate from step 4 until...



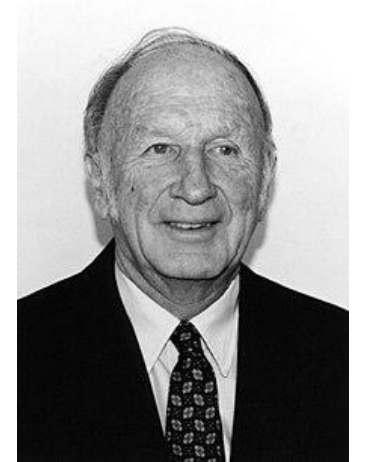
Cobweb analysis of the logistic map

- The animation shows the diagram of the **logistic map** when the value of r is changed
- Stable fixed points correspond to a decreasing «**staircase**» or an inward **spiral**, and vice versa
- A rectangle denotes a **period 2 orbit**
- Filled diagram → welcome to chaos!



Edward Norton Lorenz

- In 1950, meteorologist Ed Lorenz started building complex **mathematical models** for the **prediction of weather**
- In 1961, he developed an ODE model of the Earth's atmosphere that had 12 equations/variables: temperature, pressure, humidity, wind directions, etc.
- One day, he wanted to repeat one part of a run, but **used slightly different** initial conditions, because the computer was only printing the **first three digits** after the comma: with his big surprise, the conclusions of the model this time were **completely different**



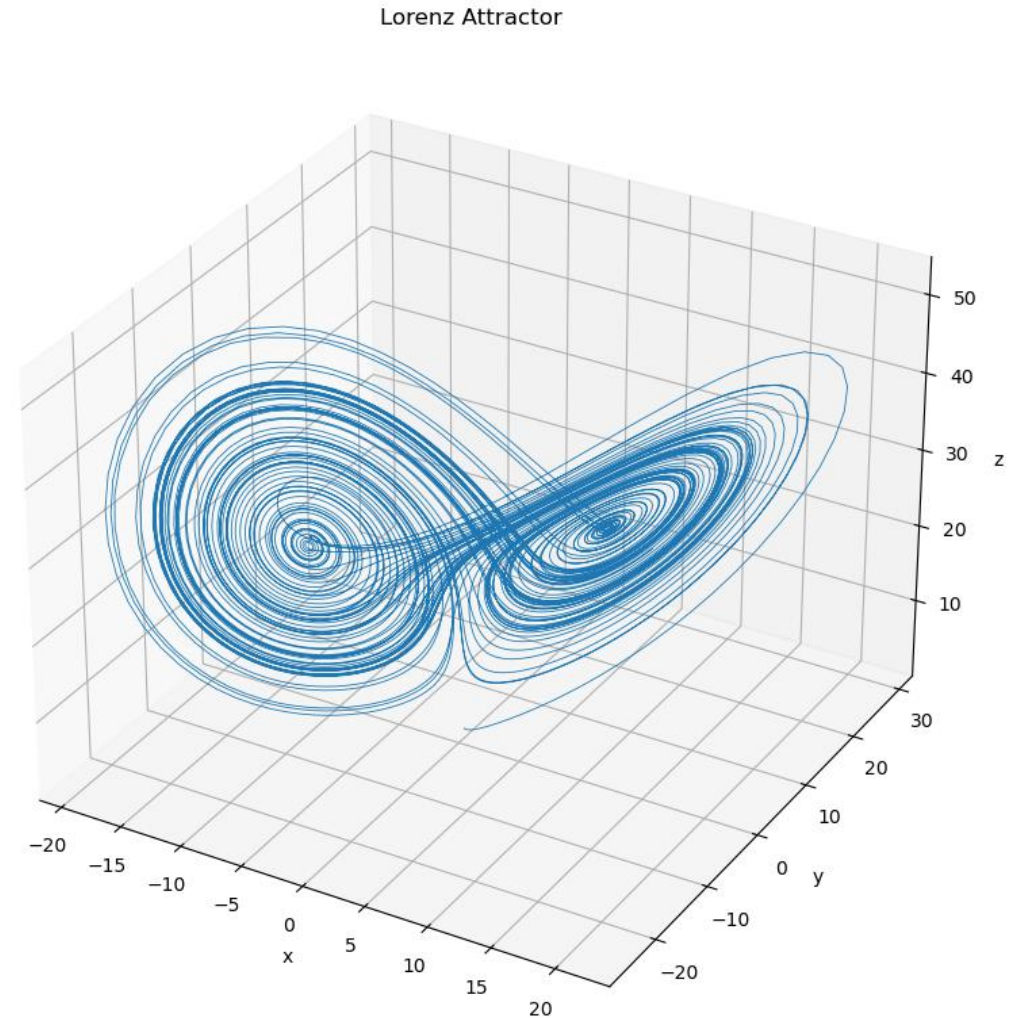
The Lorenz system

- He decided to investigate the issue, trying to **pinpoint the issue**
- So he started to **remove variables** to see whether the phenomenon was **happening again**
- Eventually (with the help of programming pioneer **Margaret Hamilton**) he found the **smallest model** that was showing such sensitiveness to initial conditions, now called the Lorenz system:

$$\begin{aligned}dX/dt &= \sigma(y - x) \\dY/dt &= x(\rho - z) - y \\dz/dt &= xy - \beta z\end{aligned}$$

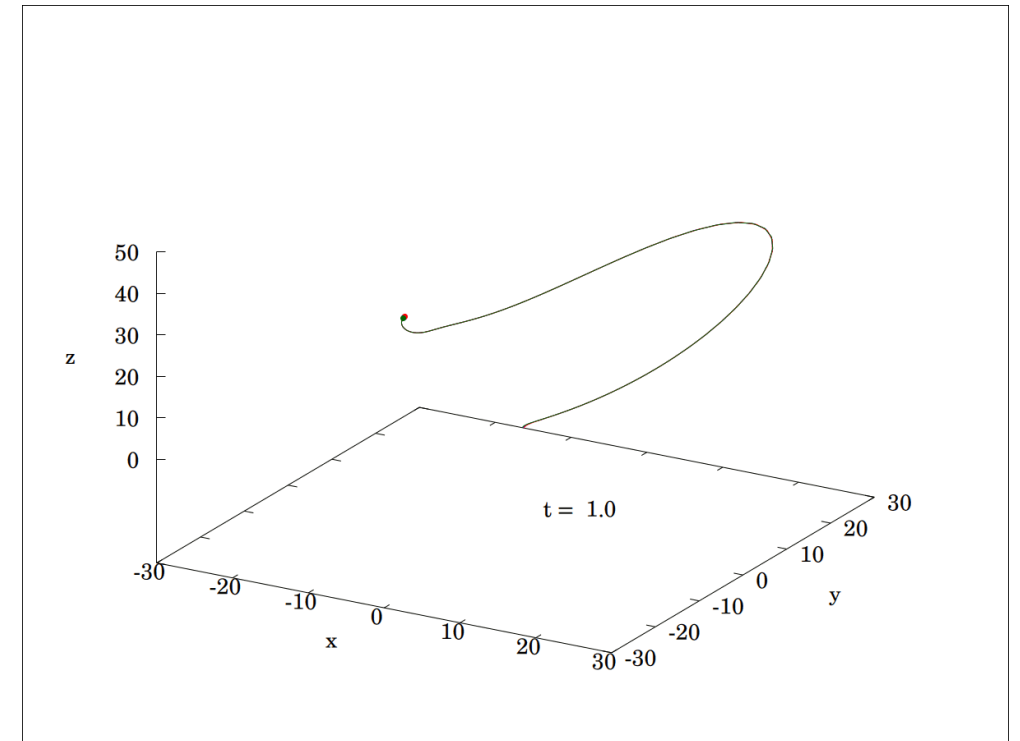
Phase diagram of the Lorenz system

- Let's simulate the **Lorenz system** using $\sigma = 10, \rho = 28, \beta = 2.667$
- Let's put the **three variables** on the respective **axis**
- The phase diagram shows that there are **two attractors**
- The dynamics is constantly changing and never repeats itself



The Lorenz system is chaotic

- Three simulations starting from **very close regions** lead to **radically different trajectories**
- This is a simple model, imagine how complex is to **predict weather!**
- This is why meteorologists often **repeat the simulations from many starting points**, close to each other, to collect and analyse a statistics

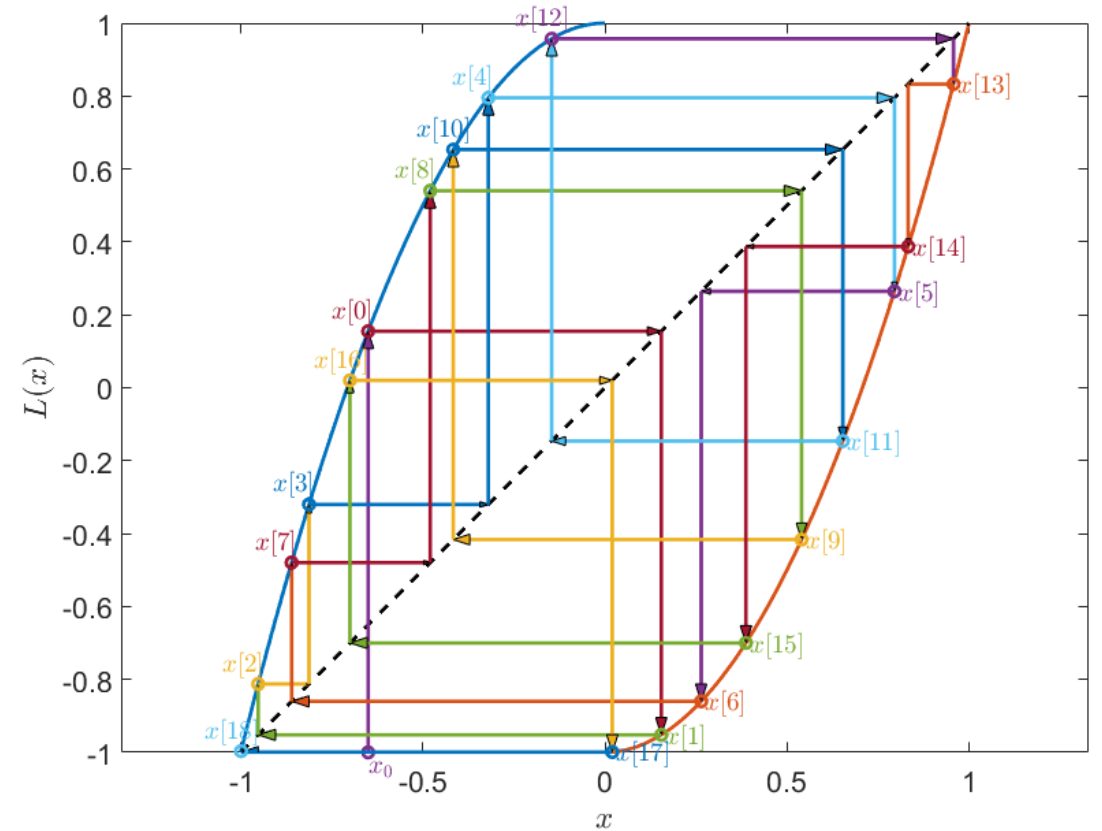


Lorenz system's Cobweb diagram

- Not easy (not a map, technically)
 - Instead consider the map

$$L(x) = \begin{cases} 1 - 2|x|^2, & x \leq 0, \\ -1 + 2x^2, & x > 0. \end{cases}$$

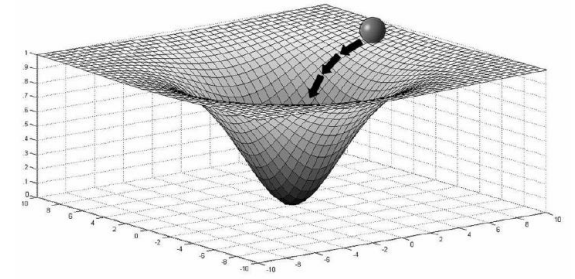
- Check [Zhang et al., Chaos Solitons Fractals 2009] for details
- The Lorenz map is **clearly chaotic**



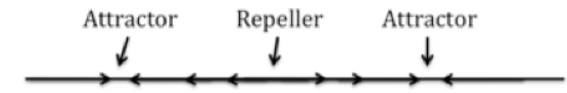
Fixed points

- A **fixed point** is a state that **does not change** when we apply the map, that is, $f(x) = x$
- There exist two types of fixed points
 - **Attractors**: the trajectory points **towards the fixed point**
 - **Repellers**: the trajectory tends to **move away from the fixed point**
- Trivial example: $x_{n+1} = x_n^2$, two fixed points: 0 and 1
 - Are they attractors or repellers?

Local and global attractors



- In dynamical systems, local and global attractors describe different scales of attraction within the system's state space



- A **local attractor** is a set of states toward which the system **tends to evolve** from initial conditions that are **close to the attractor** (i.e., fixed point)
- A **global attractor** is a set of states toward which the system tends to evolve from a wide range of initial conditions, covering the entire state space
- A **basin of attraction** is the region of the phase space such that **any point** in that region will **asymptotically converge** to the attractor



Strange attractors

Attractors: set of states toward which the **system tends to evolve**

A **strange attractor** is a **fractal** attractor

Chaotic systems tend to have **strange attractors**

Strange attractors are **locally unstable** but **globally stable**: small changes to the initial condition lead to **very different** outcomes but they **remain bounded**

Some more chaotic attractors

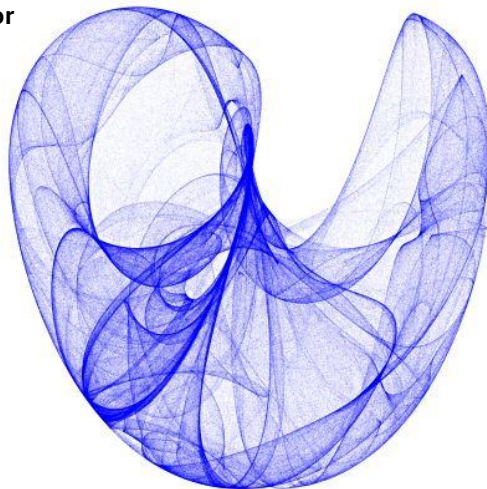


<https://www.youtube.com/watch?v=idpOunnpKTo&list=LL&index=1>

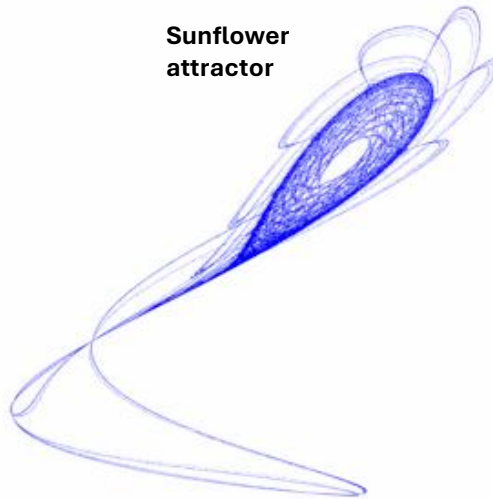
A list of chaotic equations and maps

- The author of the previous video maintains a [list of chaotic equations, ODEs and maps](#)
- As you can see in the figure on the right, you will find some **familiar faces, but also** some crazy systems (examples below)

Peter de Jong's attractor



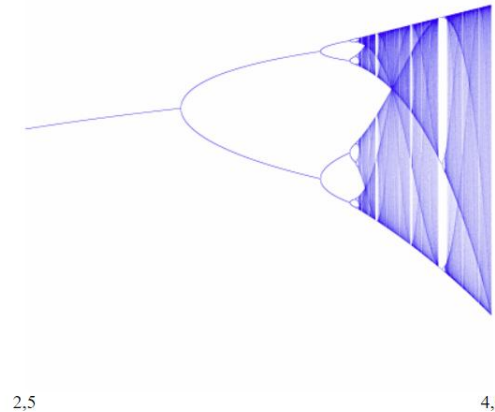
Sunflower attractor



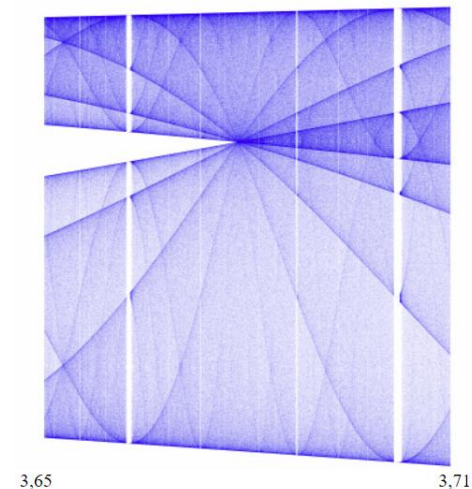
Die Logistic Map habe ich hier [\[156\]](#) gefunden.

$$x_{n+1} = ax_n(1 - x_n)$$

Das Bifurkationsdiagramm der Logistic Map ist das Bekannteste (für $a = 2,5$ bis $a = 4,0$).



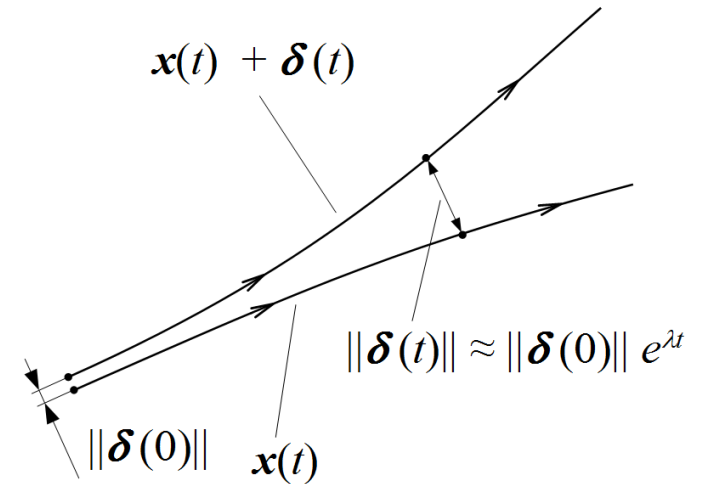
Ein Ausschnitt des Bifurkationsdiagramms für $a = 3,65$ bis $a = 3,71$.



Das Plugin zur Erzeugung des Attraktors (für Cinema Version 11) kann auf der [Download](#) Seite heruntergeladen werden.

Lyapunov exponent

- Aleksandr Lyapunov, russian mathematician
 - Defined the concept of **Lyapunov exponent**, a measure of the **rate of separation between infinitesimally close trajectories**
 - Different initial separations can have different exponents, hence the **maximum exponent** (MLE) is usually considered
- Intuitive interpretation of Lyapunov exponents λ :
 - >0 : the system is **chaotic**
 - $=0$: the system does not converge nor diverge
 - <0 : the system is **stable**
- There is a [Python library](#) to compute MLE

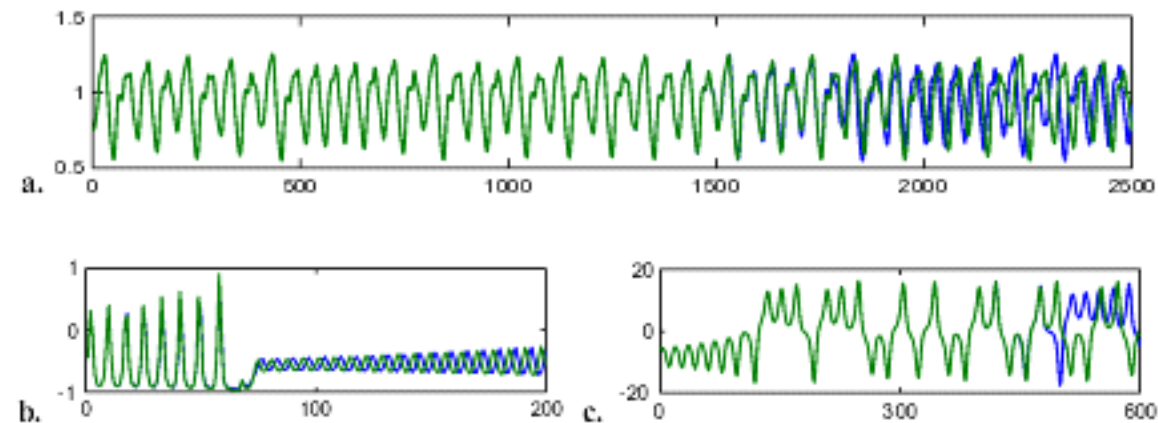


Can we exploit chaos?

- Chaotic sequences are **deterministic** and yet they tend to be **unpredictable**
- That's an excellent quality for **random numbers generation**
- You can indeed generate random numbers, one after the other, by applying a **chaotic iterative map**
See [Andrecut, Int J Modern Phys B, 1998] but also [Nobile *et al.*, IEEE CEC 2020]

Can we «learn» a chaotic sequence?

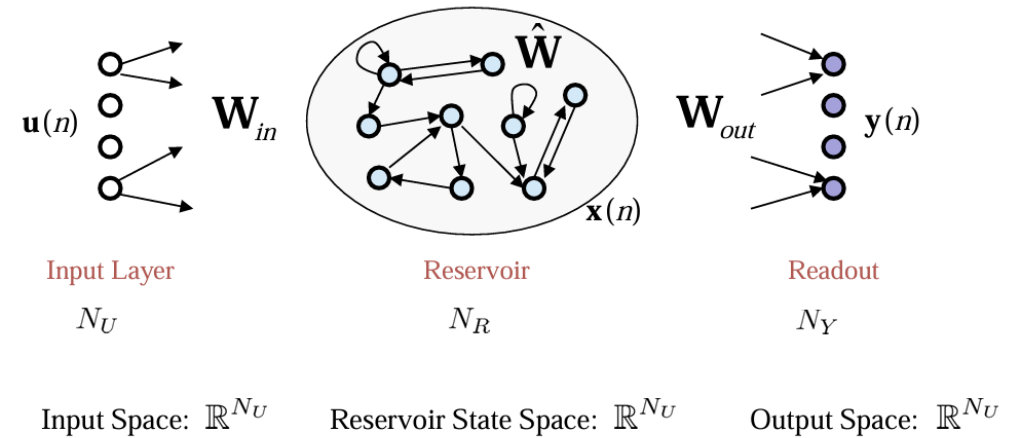
- More or less. The so-called «**liquid-state machines**» aka **Echo-State Networks** (ESN) have shown a remarkable capability in predicting (for a while...) chaotic time-series



- ESN is a class of artificial neural networks, characterized by a large amount of **randomly interconnected neurons**

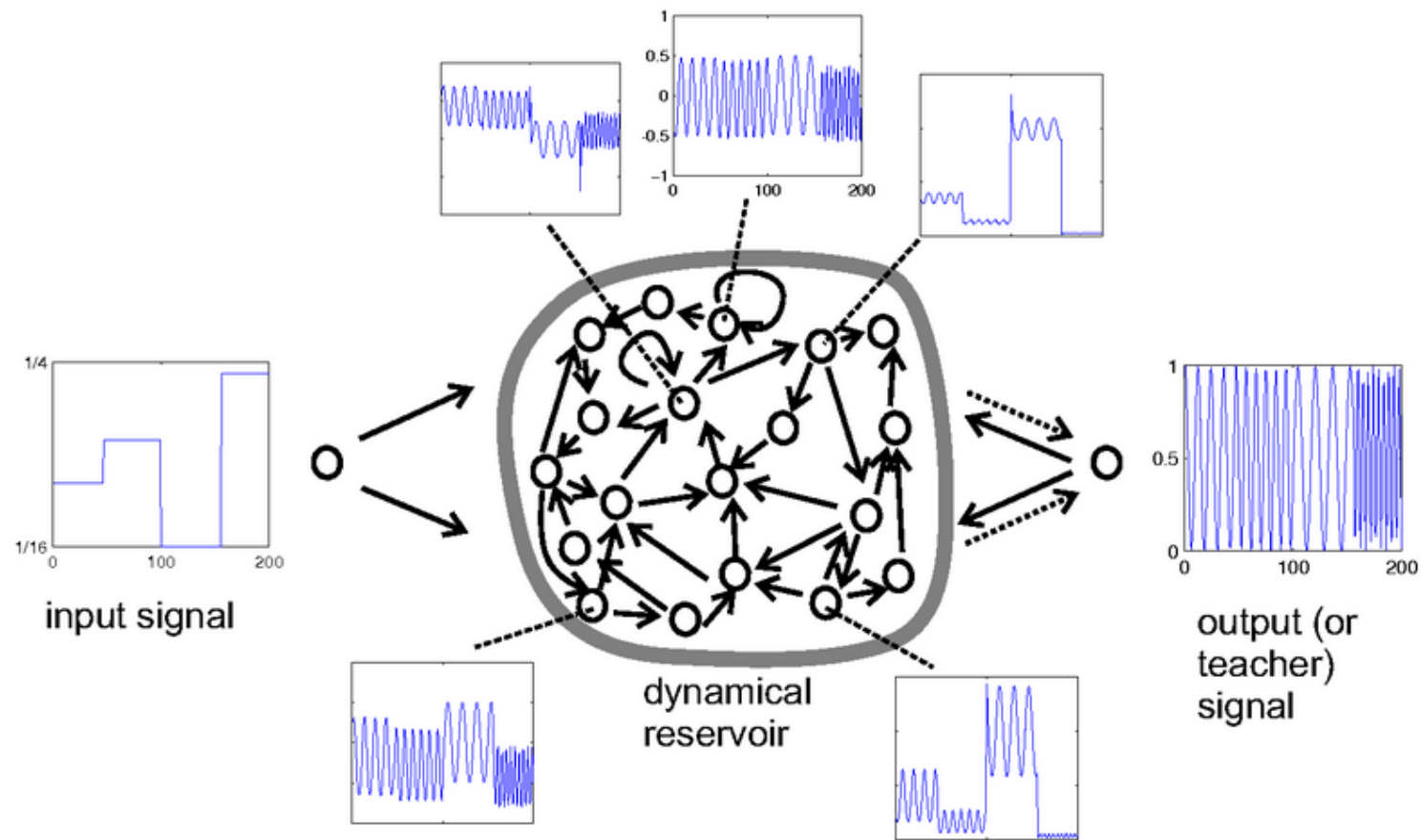
Echo-State Networks

- ESNs are composed of **three spaces**
 - The input layer
 - The reservoir space
 - The output («readout») space



- The reservoir is usually **pretty large**. It is **untrained, randomly connected, sparse, strictly non-linear**
- The **readout** layer is where the **training** happens: the weights are updated in order to minimize the prediction error

Basic schema of a ESN



Check out [\[Attiya and Parlos, IEEE Tran Neural Networks 2000\]](#)

This topic is WIDE

- The topics described in this lecture touched several disciplines
 - **Cybernetics**
 - **Systems' theory**
 - **Fractals theory**
 - Modeling and simulation
 - Mathematics
 - Theory of dynamic systems
 - Meteorology
 - Ecology and study of populations
 - Computer Science and machine learning
- The goal was **a gentle introduction to chaos**, but the whole field deserves a course on its own

Summary

- Some deterministic dynamical models can show an extreme sensitiveness to initial conditions: that is the concept of Chaos
- That prevents a faithful simulation of chaotic models: we never know the initial conditions with absolute certainty
- Phase diagrams can help in characterizing the behavior of a system
- We can still learn a chaotic sequence using Echo State Networks

